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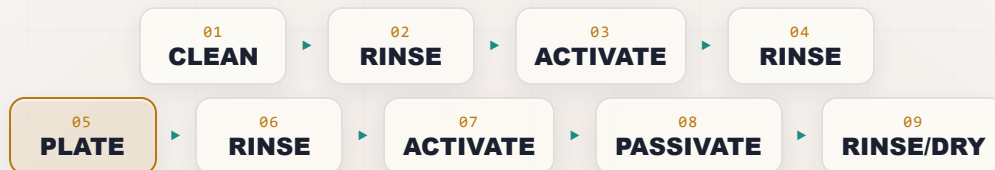
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PLATING POSTERS INC • METAL FINISHING REFERENCE SERIES

ALKALINE ZINC

THE COMPLETE TRAINING MANUAL

A full training course for the brand-new operator — from “what is an anode?” to a signed certificate of completion. Modern cyanide-free alkaline (zincate) zinc on steel. Assumes you know nothing. Teaches you everything.



OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • BEGINNER-FRIENDLY STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer specifications, current Safety Data Sheets (SDS), and applicable OSHA, EPA, REACH, and local regulations before production use.

— FIND YOUR WAY

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READ FRONT-TO-BACK THE FIRST TIME · KEEP ON THE SHELF FOREVER AFTER

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REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU ARE IN EXACTLY THE RIGHT PLACE

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — an **alkaline zinc plating line**. Maybe you started this week. Maybe you have been racking parts for a year and finally want to know *why* you do the things you do. Either way, you are in the right place, and we are glad you are here.

Here is the most important thing we can tell you up front: **this manual assumes you know nothing about plating, chemistry, or electricity**. That is not an insult. That is a promise. Every term gets defined the first time we use it. Every step explains not just *what* to do but *why* it matters. Nobody is born knowing what “throwing power” means or why a rinse tank can quietly ruin a whole shift. You learn it. We are going to teach it.

This manual is written in the spirit of the *plain-language* guides — friendly, plain-spoken, and patient. We would rather over-explain than leave you guessing in front of a tank full of hot caustic.



REMEMBER

You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: clean it, rinse it, activate it, rinse it, plate it, rinse it, bright-dip it, passivate it, and rinse/dry it. Reading it in order matters, because (as you'll soon learn) **the order of the plating line is not random** — and neither is the order of this book.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

As you read, you will run into little flagged notes. Here is what each one means.



TIP

A shortcut, a trick of the trade, or a smarter way to do something — the kind of thing a 20-year veteran would lean over and tell you. These save you time and headaches.



REMEMBER

A core idea worth locking in. If you forget everything else on the page, don't forget these. They're the load-bearing ideas.



HEADS-UP

A safety warning. Read every single one. These are the notes that keep your skin, your eyes, and your lungs intact. We use them *liberally* because an alkaline zinc line has real hazards — especially the hot, strong caustic at its heart.



TECHNICAL STUFF

Deeper chemistry or theory for the curious. Totally optional. Skip these and you can still run the line perfectly. Read them and you will understand it on a deeper level.



FUN FACT

A bit of interesting trivia to make the science stick (and to give you something to say at lunch).

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Besides the margin icons, each station chapter ends with two recurring tools. The first is a **Teach-Back** checklist (sometimes paired with a red “Top Mistakes” card) — the things you must be able to *say out loud, unprompted* before you're cleared on that station. The second is a bordered **Sign-Off** box at the very end of the station — the short statement your *trainer initials* (with you) once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you have to pass through.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

THE WHOLE FOUNDATION — MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what electroplating is** in plain language — how electricity moves metal from one place onto another.
2. **Describe what alkaline zinc plating does** and why a shop chooses it — even coverage, ductile deposits, and sacrificial corrosion protection on steel, with no cyanide in the modern version.
3. **Name all nine stations of the line in order** and explain *why* the order cannot be rearranged.
4. **Understand what each tank is for** — what goes in, what comes out, and what “good” looks like at every handoff.
5. **Recognize the main hazards** at each station — hot caustic, mineral acids, electrical, mist, and chromate chemistry — and know the correct PPE and emergency response for each.
6. **Use the basic field tests** every operator relies on, like the water-break test and conductivity checks.
7. **Spot trouble early** — read a deposit and know whether the problem is upstream, in the bath, or in your handling.
8. **Talk the language** — use the right words so you can communicate clearly with your lead, your chemist, and your chemistry supplier.



REMEMBER

You are not expected to hit all of these on day one. Plating is a craft. The goal is steady competence, not instant mastery.

• THE NINE-STATION PROCESS AT A GLANCE

Here is the whole alkaline zinc line in one strip. Don't memorize it yet — just get a feel for the rhythm: **Clean** → **Rinse** → **Activate** → **Rinse** → **Plate** → **Rinse** → **Activate** → **Passivate** → **Rinse/Dry**.



REMEMBER

Notice the pattern: **a rinse follows almost every working tank**. Rinses are not breaks — they are checkpoints that keep one chemistry from poisoning the next. Every working chemistry wants to “hitch a ride” into the next tank on the surface of your part, and rinses are what stop that. They aren't filler between the “real” steps; they're some of the most important steps you'll perform.

CHAPTER 1

WHAT IS ELECTROPLATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT PLATING IS

THE ONE-SENTENCE VERSION

Electroplating is using electricity to coat one metal with a thin layer of another metal. That's the whole idea. Now let's unpack it so it actually means something.

Imagine you have a steel bolt. Steel is strong and cheap, but it rusts. You want to give that bolt a thin protective skin of **zinc** — a metal that protects steel from rust (we'll explain exactly how in a minute). You *could* try to paint zinc on, but that would be uneven and weak. Instead, we use electricity to grow a perfectly even, tightly-bonded layer of zinc directly onto the steel, atom by atom. That process is electroplating.

HOW ELECTRICITY MOVES METAL: THE KITCHEN-SINK PICTURE

Picture a tank full of liquid called the **electrolyte** (or just “the bath”). This liquid has zinc dissolved in it — invisible zinc, broken down into tiny charged particles. In *acid* zinc the zinc floats around as simple positive zinc ions (Zn^{2+}). In *alkaline* zinc, the story is a little different and worth knowing now: the zinc is dissolved in **caustic (sodium hydroxide)**, and it lives in solution as a **negatively-charged zincate ion** — written $\text{Zn}(\text{OH})_4^{2-}$. Same zinc metal, very different chemical “wrapper.” That wrapper is a big part of why alkaline zinc behaves the way it does.

You hook two things up to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction “DC” current plating needs):

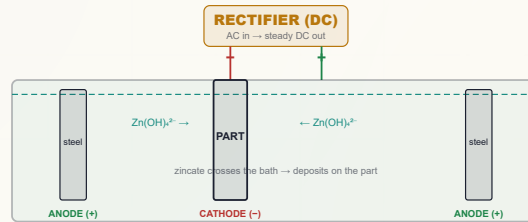
- **The cathode** — this is your *part* (your steel bolt), connected to the **negative** terminal. Just remember “cathode = the part, the thing collecting metal.”
- **The anode** — bars or plates hanging in the same tank, connected to the **positive** terminal.

Here's a key difference from acid zinc that we'll come back to often: **in modern alkaline zinc, the anodes are usually plain STEEL, not zinc.** They don't dissolve to feed the bath. Instead, the zinc metal is replenished separately, by dissolving zinc in a side “dissolver” tank or by periodic chemical additions. (Some shops still run *zinc* anodes, but steel anodes are the modern norm — more on this at Station 05.)

When you turn on the rectifier:

1. Current flows. The zincate ions in the bath are driven toward the negatively-charged part (the cathode).
2. When a zincate ion reaches the part's surface, it gives up its caustic “wrapper,” grabs electrons, and turns back into solid zinc metal, sticking to the surface.
3. At the same time, because the bath is high-efficiency-*poor* (we'll explain), a good chunk of the current also splits water into **hydrogen gas** at the part and **oxygen** at the steel anodes.

THE ELECTROPLATING CELL (ALKALINE ZINC)



★ REMEMBER

Three words to lock in now: **Cathode = the part** (collects zinc). **Anode = the (usually steel) bars**. **Electrolyte = the bath** (the caustic liquid that carries zinc as zincate). Almost everything in plating is a variation on this one idea.

⚙️ TECHNICAL STUFF

At the part, a **reduction**: $\text{Zn(OH)}_4^{2-} + 2\text{e}^- \rightarrow \text{Zn}^0 + 4\text{OH}^-$. Because the deposit comes from a complexed (zincate) ion rather than a free Zn^{2+} ion, it deposits more slowly and more *evenly* — the chemistry that gives alkaline zinc its famous throwing power. With *steel* anodes the anode reaction is mostly oxygen evolution ($4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$), which is exactly why the metal must be fed in separately.

💡 FUN FACT

"Galvanize" and the unit "volt" trace back to Luigi Galvani and Alessandro Volta, two 1790s Italian scientists who argued over whether electricity came from frog legs or from metals. Volta won — and you'll read volts on your rectifier every day.

⚠️ HEADS-UP

Plating rectifiers push enormous current — hundreds to thousands of amps — low voltage but very high current, into a tank full of conductive liquid. **Never reach into an energized tank, and never do maintenance until equipment is locked and tagged out (LOTO).** Electricity plus hot caustic plus your hand demands total respect.

CHAPTER 2

WHAT IS “ALKALINE ZINC”?

AND WHY DO WE USE IT — EVEN, DUCTILE, SACRIFICIAL PROTECTION ON STEEL, CYANIDE-FREE

ZINC PLATING COMES IN FAMILIES

“Zinc plating” just means coating something with zinc. But there are *several different recipes* for the bath that does it. The two big families are:

- **Acid zinc** — a bath that is chemically *acidic* (low pH, around 4.8–5.8). Fast and bright, but covers recesses poorly.
- **Alkaline zinc** — a bath that is chemically *basic* (high pH, soapy-feeling). This manual is about alkaline zinc.

And alkaline zinc itself comes in two sub-families:

- **Cyanide alkaline zinc** — the old way. Zinc dissolved in sodium cyanide and caustic. It plates beautifully but uses **cyanide**, which is acutely lethal and a major waste-treatment and safety burden.
- **Cyanide-free (zincate) alkaline zinc** — the modern way, and **what this manual covers**. Zinc dissolved in **sodium hydroxide (caustic) only**, with organic additives doing the work cyanide used to do. No cyanide at all.



TECHNICAL STUFF

pH is the scale we use to measure how acidic or basic a liquid is. It runs from 0 (strong acid) to 14 (strong base), with 7 being neutral (pure water). Higher number = more basic/caustic. Our alkaline zinc bath runs at extremely high pH — effectively **>14** because of the high free-caustic load. That is far more caustic than household lye water. This is the single most important safety fact about this line: **the plating bath itself is a hot, strong caustic**.

More precisely, our process is **cyanide-free alkaline zincate**. The supporting chemistry is **sodium hydroxide (NaOH, “caustic soda”)**, which both dissolves the zinc into zincate and carries the current. A small set of **purifiers, carriers, and brighteners** (organic additives) controls grain, brightness, and the all-important even distribution.

THE BATH AT A GLANCE

COMPONENT	TYPICAL RANGE	WHAT IT DOES
Zinc metal (as zincate)	6–15 g/L (0.8–2.0 oz/gal)	The metal you’re plating
Sodium hydroxide (NaOH)	90–150 g/L (12–20 oz/gal)	Dissolves zinc as zincate; carries current
NaOH : Zn ratio	~8:1 to 12:1 (target ~10:1)	Master control for distribution & efficiency
Sodium carbonate (Na ₂ CO ₃)	Builds up; treat at >60–90 g/L	A contaminant byproduct, not an addition
Purifier	Per supplier’s TDS	Sequesters tramp metals; prevents dark/skip plate
Carrier (conditioner)	Per supplier’s TDS	Refines grain, improves coverage
Brightener	Per supplier’s TDS	Adds shine and leveling



REMEMBER

The number that rules this bath is not zinc *or* caustic alone — it’s the **ratio between them (NaOH:Zn, target ~10:1)**. Get that ratio right and the bath throws beautifully and behaves; let it drift and you get poor distribution, low efficiency, and burning. We’ll come back to this ratio again and again.



TIP — ACID ZINC VS. ALKALINE ZINC, THE HEADLINE CONTRAST

Notice the zinc concentration: alkaline zinc carries only about **6–15 g/L** of zinc, versus acid zinc’s **15–45 g/L**. *Less metal, dissolved as a complex, deposited slowly and evenly* is the whole secret to alkaline zinc’s superpower — uniform coverage. Acid zinc loads up on cheap metal and plates fast but unevenly. Two opposite design philosophies.

• WHY DO SHOPS USE ALKALINE ZINC?

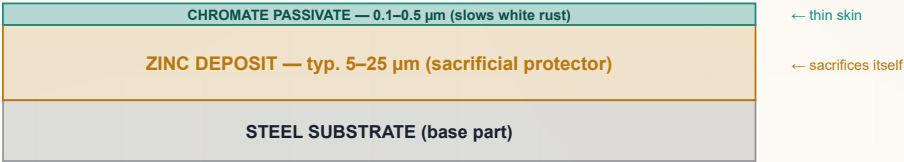
Three big reasons: **even coverage** (throwing power), **ductile, well-distributed deposits**, and — in the modern version — **no cyanide**. Let's take them one at a time.

REASON 1: SACRIFICIAL CORROSION PROTECTION

Steel rusts. Rust (iron oxide) is what happens when iron reacts with oxygen and moisture. Zinc protects steel in a clever way. It's not just a physical barrier (though it is that too). Zinc is what we call a **sacrificial coating**. Zinc is *more chemically reactive* than steel — it “wants” to corrode more eagerly than iron does. So when a zinc-coated steel part gets wet and starts to corrode, the **zinc corrodes first and protects the steel underneath**. Even if the coating gets scratched and bare steel is exposed, the surrounding zinc keeps protecting that scratched spot.

THE FINISH, LAYER BY LAYER (NOT TO SCALE)

ZINC PROTECTS STEEL • PASSIVATE PROTECTS ZINC



TECHNICAL STUFF

This is called **galvanic** or **cathodic protection**. In an electrochemical pairing, the more *active* metal becomes the **anode** and corrodes, while the less active metal becomes the **cathode** and is protected. Zinc is more active than iron, so zinc takes the hit. This is why a scratch in zinc plating doesn't immediately rust the way a scratch in paint would — the zinc keeps “throwing itself on the grenade” for the exposed steel nearby.



FUN FACT

Same principle as the chunky “sacrificial anodes” bolted to boat hulls and water heaters — blocks of zinc or magnesium that corrode away so the steel doesn't. Your plated bolt carries a microscopic version of the same bodyguard.



REMEMBER

Zinc protects steel by sacrificing itself. That single idea is the entire reason this line exists. Everything you do on this line is in service of laying down a good, even layer of that sacrificial zinc — and alkaline zinc is especially good at the *even* part.

HOW THICK SHOULD THE ZINC BE? (SERVICE CONDITIONS)

How much zinc a part needs depends on how harsh a life it'll lead. The industry standard **ASTM B633** sorts parts into four **Service Conditions (SC)**. The harsher the environment, the thicker the minimum zinc. Typical commercial zinc runs **5–15 µm**; severe-service parts go up to **25 µm**.

SERVICE CONDITION	ENVIRONMENT	MIN. ZINC THICKNESS
SC 1 — Mild	Indoor, dry (warm rooms, sheltered)	5 µm
SC 2 — Moderate	Indoor, some condensation/humidity	8 µm
SC 3 — Severe	Outdoor, moderate weather exposure	12 µm
SC 4 — Very Severe	Outdoor harsh / marine / road salt	25 µm



TIP

You don't pick the Service Condition — the customer's spec or print does. Your job is to plate long enough to hit the minimum thickness for the SC on the job ticket. With alkaline zinc, the good news is that thickness comes out *evenly* across the part, so you're less likely to be thin in a recess. The flip side: it plates **slower**, so hitting thickness takes more time than acid zinc would.

REASON 2: OUTSTANDING THROWING POWER & EVEN DISTRIBUTION

This is where alkaline zinc *shines* — and it's the mirror image of acid zinc's biggest weakness. “**Throwing power**” is the bath's ability to plate evenly into deep recesses, blind holes, the insides of tubes, and the low spots on a complex part. Alkaline zinc has **excellent** throwing power. The zincate complex naturally “evens out” the deposit, plating low-current recesses nearly as fast as high-current edges.



TECHNICAL STUFF

The shop numbers you'll hear are **surface-to-recess thickness ratios**. Alkaline zinc runs roughly **1.5:1 to 2:1**, versus acid zinc's **3:1 to 5:1**. A number *closer to 1:1* means *more even coverage* — so alkaline's smaller spread is the better one. The cause is the complexed zincate ion plus a naturally lower cathode efficiency that *self-levels*: edges that try to plate too fast just waste extra current on hydrogen instead, letting the recesses catch up.

REASON 3: DUCTILE DEPOSITS, AND NO CYANIDE

Alkaline zinc deposits tend to be **fine-grained, ductile, and low-stress** — they bend and form without cracking or flaking, which matters for parts that get crimped, threaded, or formed after plating. And the modern bath uses **no cyanide**, dramatically simplifying both worker safety and waste treatment compared to the old cyanide zinc.



FUN FACT

For decades, “alkaline zinc” *meant* “cyanide zinc,” and shops accepted the cyanide as the price of good throwing power. The development of organic additive packages that reproduce that throwing power *without* cyanide, starting in the 1960s–70s and maturing since, is one of the great quiet wins of modern plating chemistry.

THE HONEST TRADEOFFS

A good operator knows the weaknesses too. Alkaline zinc is not the answer for everything:

- **Slower plating / lower efficiency.** Cathode efficiency is only about **50–80%** (versus acid zinc's 95–98%). A lot of current goes into making hydrogen gas instead of zinc. That means **alkaline zinc plates 2–3× slower** than acid zinc at the same current.
- **Hotter caustic safety burden.** The bath is a strong, often warm caustic — a more serious skin/eye hazard than acid zinc's mild bath.
- **Carbonate buildup.** Caustic baths slowly absorb CO₂ from the air and form **sodium carbonate**, which accumulates and eventually must be removed.
- **Anode handling is different.** Steel anodes don't replenish zinc, so metal and ratio must be actively managed — get it wrong and the bath drifts.
- **Hydrogen embrittlement risk on hard steel** — present here just as in acid zinc, requiring a relief bake on high-strength parts (covered later).



TIP

If you ever hear “this part needs to plate evenly into a deep blind hole” or “this part gets formed/crimped after plating,” your first thought should be *alkaline zinc* — its throwing power and ductility are exactly why it was chosen. If you hear “we need maximum parts per hour, simple geometry,” that's acid-zinc territory. Process selection is about matching the bath to the part.

A WORD ABOUT THE FINAL STATIONS: BRIGHT DIP + PASSIVATION

The alkaline zinc line ends with an **acid bright dip (activation)** followed by **trivalent chromate passivation**. After plating, you have zinc — but it comes out of a caustic bath, so it carries a faint film and is slightly dull. A quick **dilute nitric (or other mineral) acid dip** neutralizes the caustic film, brightens the zinc, and chemically “prepares” the surface so the chromate grabs evenly. Then the **chromate conversion coating** reacts with the zinc to grow a thin, tough film that dramatically slows **white rust** and can add color — **clear/blue, yellow, or black**. Modern passivates use **trivalent chromium**, Cr(III), the RoHS-compliant, far less hazardous form.



HEADS-UP

The old generation of passivates used **hexavalent chromium — Cr(VI)** — a confirmed **human carcinogen**, heavily restricted. Most modern shops use trivalent. **Know which one your shop uses.** Any hexavalent chromate still in use carries strict OSHA requirements (PEL 5 µg/m³, supplied-air respirators, medical surveillance per 29 CFR 1910.1026) that go far beyond ordinary line PPE.



REMEMBER

Zinc without passivate is a coating waiting to white rust. Passivation isn't an optional finishing touch — it's what makes the zinc layer actually last in service.

CHAPTER 3

HOW A PLATING LINE WORKS

THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A LINE IS A SEQUENCE OF TANKS

An alkaline zinc line is not one tank. It's a **sequence of tanks** that a part travels through in a strict order. Think of it like a kitchen line or a car wash: each station does one job, and each job prepares the surface for the next. Parts ride down the line in one of two ways:

- **Rack plating** — Parts are individually hung on a **rack** (a frame with hooks or clips that holds parts and carries electrical current to them). Used for larger or delicate parts.
- **Barrel plating** — Lots of small parts (screws, washers, small fittings) are dumped together into a perforated rotating **barrel** that tumbles them through the chemistry. **Alkaline zinc is especially popular for barrel work**, because its great throwing power evens out the deposit even when parts are buried in a tumbling mass and only see current intermittently.



TIP

You'll see different numbers for rack vs. barrel — barrel baths often run a touch lower in zinc and tweak the additive package, and barrel current densities are much lower because tumbling parts share the current. When in doubt about which numbers apply, check whether you're running rack or barrel and read the matching column on your shop-floor poster.

• THE WHOLE ALKALINE ZINC SEQUENCE: 9 STATIONS

#	STATION	THE ONE THING IT DOES	TEMP	TIME
01	Alkaline Soak Clean	Removes all oil, grease, and shop soil	130–160°F	3–10 min
02	Rinse (Pre-Activation)	Washes off cleaner before the acid dip	Ambient	30–60 s
03	Acid Activation / Pickle	Dissolves surface oxide; “wakes up” the steel	Amb–110°F	15–60 s
04	Rinse (Pre-Plate)	Guards the plating bath from acid & iron drag-in	Ambient	30–60 s
05	Alkaline Zinc Plating	Deposits the even zinc layer	70–95°F	10–60 min
06	Rinse (Post-Plate, recovery)	Washes caustic off; often a drag-out recovery tank	Ambient	30–60 s
07	Acid Bright Dip / Activate	Neutralizes caustic film; brightens; preps for chromate	Ambient	5–20 s
08	Chromate / Passivate	Adds a chromate skin for extra corrosion protection	70–110°F	30–90 s
09	Final Rinse / Dry	Cleans up and dries without damaging the coating	Amb–150°F	30–60 s + dry

The basic rhythm: **Clean → Rinse → Activate → Rinse → Plate → Rinse → Activate → Passivate → Rinse/Dry.**

• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS



REMEMBER

Every station either protects the next station or poisons it. There is no neutral step. A part that leaves the cleaner still carrying a film of cleaner contaminates the acid. A part that leaves the acid still wet with acid drags acid and iron toward the plating bath. What you do (or fail to do) at Station 1 shows up as a defect at Station 5. The line is a chain, and a chain is only as strong as its weakest link.

This is also why **the plating bath itself is precious**. Unlike the cleaner and the acid (dumped and remade regularly), a well-maintained alkaline zinc bath runs a long time. Here's the good news that's *different* from acid zinc: alkaline zinc is **far more forgiving of iron contamination**. Iron that drags in quietly **precipitates out as iron hydroxide and gets caught by the filter** — the bath largely cleans itself of iron. Its enemies are different: **carbonate buildup, organic breakdown, drifting NaOH:Zn ratio, and certain tramp metals** the purifier has to chase.



TIP

Think of the alkaline zinc tank as the crown jewel of the line. Everything *before* it (clean, rinse, activate, rinse) exists to hand it a perfectly clean, bare, well-rinsed part. Everything *after* it (rinse, bright dip, passivate, rinse/dry) exists to protect and finish what the zinc created. The whole line orbits that one tank.

CHAPTER 4

WHY THE ORDER MATTERS

WALKING THE LINE — EACH STATION SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

You might think the order is just convention — that you could shuffle steps or skip one to save time. You can't. Each station exists *because of* the one before and after it. Here is the logic, link by link.

STATION 1 — CLEAN: GET RID OF THE JUNK FIRST

Plating, acid, and chromate all need to touch *bare metal*. Oil and shop soil block them. **Plate over dirt and you've just plated dirt** — the zinc bonds to the grease, not the steel, and peels off later. The free, foolproof cleanliness test is the **water-break test** (see the cleaning chapter): water sheets off a clean part in an unbroken film; on a dirty part it beads up.

**TIP**

The simplest, most powerful quality check in the whole shop is the **water-break test**, and it's free. A smooth, continuous, unbroken sheet means the surface is clean. If water beads or "breaks" into droplets, there's still oil there — *stop and re-clean*.

**HEADS-UP**

The cleaner is **hot (130–160°F)** and **strongly alkaline (pH 11–13.5)** — it causes chemical burns and can seriously injure your eyes. Wear full PPE, and know where your shower and eyewash are before you need them.

STATION 2 — RINSE (PRE-ACTIVATION): DON'T CARRY THE CLEANER FORWARD

The cleaner is alkaline (basic). The next tank is acid. If you carry cleaner straight into the acid, it neutralizes the acid, wastes it, and weakens the activation. The rinse strips the cleaner film off first.

**REMEMBER**

"Drag cleaner into acid and you waste acid." Rinses exist to keep each chemistry from invading the next tank. The cleaner-to-acid rinse is your first firewall on the line.

STATION 3 — ACID ACTIVATION: WAKE UP THE METAL

Even a clean steel surface grows an invisible layer of **oxide** (a tarnish-like film from reacting with air). Zinc can't bond to oxide. The acid dip dissolves that oxide and leaves a chemically "fresh," active steel surface for the zinc to grip.

- **Under-dip** (too short/weak) = oxide remains = zinc peels off later (adhesion failure).
- **Over-dip** (too long/strong/hot) = you start etching and pitting the part itself, and load it with hydrogen.

**REMEMBER**

No activation, no adhesion. Period. Activation has to come right before plating, with only a rinse in between — freshly activated bare metal just grows a new oxide skin if it sits.

**HEADS-UP**

The acid dip uses **strong mineral acid** — typically HCl at 10–50% or H₂SO₄ at 5–25%. These cause severe burns and eye damage, and HCl gives off corrosive, choking fumes even at room temperature. The acid eating into metal also releases **flammable hydrogen gas**. Full acid PPE, good ventilation, and **always add acid to water, never water to acid**.

**TECHNICAL STUFF**

Hydrogen embrittlement is a real, serious risk. During pickling and plating, atomic hydrogen can diffuse into steel's crystal structure. In high-strength steel (above ~40 HRC, or 180 ksi tensile), trapped hydrogen can cause catastrophic cracking later, under load — sometimes hours or days after the part is in service. The fix is **baking**: 375 ±25°F for at least 8 hours, started **within 4 hours** of the last plating step (ASTM B850). For prone parts, use *inhibited* acids and minimum dip time. A failed bolt or structural part can hurt people — always check the customer spec.

STATION 4 — RINSE (PRE-PLATE): THE GUARD RINSE

This is the **guard rinse** protecting your most valuable tank. Acid dragged into the plating bath neutralizes precious free caustic, can introduce chloride/sulfate that the bath doesn't want, and carries dissolved **iron** from the pickle into the zinc bath.

**REMEMBER**

Alkaline zinc is more forgiving of iron than acid zinc — its filter catches precipitated iron — but **don't get lazy**. Dragging acid into a caustic bath wastes caustic (neutralization), upsets the NaOH:Zn ratio, and floods the filter with iron sludge. Keep this rinse tight: target < 200 µS/cm, double rinse strongly recommended.

**TECHNICAL STUFF**

The recurring villain that justifies all those rinses is **drag-out** (and its mirror image, **drag-in**). Drag-out is the film of solution clinging to a part as it leaves a tank; carried into the next tank it becomes drag-in — contamination. Operators measure rinse cleanliness with **conductivity**, in microsiemens per centimeter (µS/cm) — basically how many dissolved ions are in the water. *Higher conductivity = more dragged-in chemistry = a dirtier rinse*. Critical rinses on this line target **under 200 µS/cm**.

STATION 5 — ALKALINE ZINC PLATING: THE MAIN EVENT

Now — and only now, with a perfectly clean, freshly activated, well-rinsed part — does the actual zinc plating happen. The bath is high-pH caustic zincate chemistry running near room temperature (70–95°F). With **steel anodes**, the zinc metal is fed separately and the NaOH:Zn ratio is actively managed.

**HEADS-UP**

The plating bath is **hot, strong caustic** and air agitation throws up a fine **caustic mist** you must not breathe — local exhaust ventilation is required. The rectifier runs **high amperage** — shock and arc-flash risk. Because the bath also gasses hydrogen heavily at the part (low efficiency), hard steel still absorbs enough hydrogen to require a relief bake above 40 HRC. LOTO before any rectifier work.

STATION 6 — RINSE (POST-PLATE, RECOVERY): GET THE CAUSTIC OFF

The plating bath is loaded with **caustic**. Drag caustic forward and it neutralizes your acid bright dip and contaminates the chromate. Many shops make the first post-plate rinse a **drag-out recovery (dead) rinse** — a still tank whose mildly-caustic, zinc-bearing water is periodically pumped back to top off the plating bath, saving chemistry and reducing waste.

**REMEMBER**

Caustic into the bright dip and chromate is wasteful and damaging. This rinse is the wall between the high-pH plating bath and the low-pH finishing tanks. It's also the decision point for the hydrogen-embrittlement bake on hardened steel.

STATION 7 — ACID BRIGHT DIP / ACTIVATION: WAKE THE ZINC UP

Zinc leaving a caustic bath carries a faint film and is a little dull. A quick dip in **dilute mineral acid** (commonly 0.5–1% nitric acid, sometimes dilute sulfuric) **neutralizes the leftover caustic, dissolves the dull film, and brightens the zinc** so the chromate forms an even, well-colored coating.



REMEMBER

Think of the bright dip as “activation, round two” — but this time you’re activating the *zinc* for the chromate, not the *steel* for plating. A weak or skipped bright dip is a top cause of dull, patchy, poorly-colored chromate on alkaline zinc.

STATION 8 — PASSIVATE: LOCK IN THE FINISH

Bare fresh zinc forms ugly **white rust** quickly on its own. The passivate (a chromate conversion coating) adds a thin protective skin that dramatically slows white rust and can add color — clear/blue, yellow, or black. Modern lines use trivalent chromium, Cr(III).



HEADS-UP

Even trivalent passivate is **acidic (pH ~1.5–3.5)** and causes burns. If any **hexavalent chromium (Cr⁶⁺)** is in use, it is a **confirmed human carcinogen** (OSHA PEL 5 µg/m³) requiring supplied-air respirators and medical surveillance under OSHA 29 CFR 1910.1026. Know which chemistry your shop runs.

STAGE 9 — FINAL RINSE / DRY: SEND IT OUT RIGHT

A last clean rinse removes passivate residue, and a *gentle* dry (not too hot) finishes the part without cracking the fresh chromate film. Trivalent films crack above ~150°F, so don’t over-dry.



REMEMBER

The whole nine-station sequence is one continuous logic chain: *clean it so zinc can stick* → *rinse so the cleaner doesn’t ruin the acid* → *activate so the metal is bare and ready* → *rinse so the acid and iron don’t burden the precious zinc bath* → *plate the even zinc* → *rinse so caustic doesn’t ruin the finishing tanks* → *bright-dip so the zinc takes chromate evenly* → *passivate to protect the zinc* → *final rinse and gentle dry so it ships flawless*. Every step earns its place.



TIP

When you understand *why* each station exists, troubleshooting becomes logical instead of mysterious. Peeling deposit? Look upstream at cleaning and activation. Dull, patchy chromate? Look at the bright dip and post-plate rinse. The defect almost always points back to the station that was supposed to prepare for it.

CHAPTER 5

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND CHEMICALS THAT BURN, MIST THAT HARMS, SPLASHES THAT BLIND

An alkaline zinc line works with hot caustic (both in the cleaner *and* the plating bath), strong mineral acids, electrical current at high amperage, chemical mists, and chromium chemistry. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

 **REMEMBER**
There is no part, no order, and no deadline worth an injury. Scrap can be reworked. Your eyes cannot.

• THE MASTER PPE SET (WEAR AT ALL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; a splash finds the gaps.
- **Face shield** — over the goggles for any work involving acids, hot caustic (cleaner *and* plating bath), or passivate.
- **Chemical-resistant gloves** — elbow-length, rated for both acids and alkalis (nitrile, neoprene, or PVC per the chemistry). Inspect for pinholes before each use.
- **Chemical-resistant apron** — full-length, chest to below the knee.
- **Chemical-resistant boots** — closed, non-slip, splash-proof.
- **Long sleeves / appropriate clothing** — no exposed skin where splashes are likely.

 **HEADS-UP**
Respiratory protection is required when ventilation is inadequate or when working over fuming/misting tanks (acid pickle, plating bath caustic mist, passivate). Use the cartridge or supplied-air respirator your shop specifies. If you can smell acid fumes strongly, the ventilation isn't keeping up — stop and report it.

 **TIP**
Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. Rinse your gloves *before* you take them off.

 **HEADS-UP — CAUSTIC-SPECIFIC DANGER**
Caustic (the cleaner and the plating bath) is in some ways *more* insidious than acid: it doesn't sting as sharply on first contact, so people don't rinse it off fast enough — but it keeps eating into skin and is **devastating to the eyes**. Treat any caustic contact as an emergency: flush immediately, a full 15 minutes, and get medical attention. Don't wait for it to “start hurting.”

• STATION-BY-STATION HAZARD SUMMARY

STATION	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Soak Clean	Hot alkaline (caustic), 130–160°F	Chemical burns; hot liquid splashes
02 Rinse	Residual alkaline cleaner	Mild skin/eye irritant; wet/slip hazard
03 Acid Activation	Strong mineral acid; HCl fumes; flammable H ₂ gas	Severe burns & eye damage; corrosive fumes; fire/explosion risk
04 Rinse	Dilute acid residue	Mild irritant; slip hazard
05 Alkaline Zinc Plating	Hot strong caustic bath; caustic mist; high-amperage rectifier	Severe burns; respiratory irritation; shock & arc-flash risk
06 Rinse (recovery)	Residual caustic + dissolved zinc	Caustic irritant; slip hazard
07 Acid Bright Dip	Dilute mineral acid (often nitric); NO _x fumes if nitric	Burns; nitric gives off irritating brown fumes
08 Passivate	Acidic chromate (Cr(III) pH 1.5–3.5; or Cr(VI) where still used)	Burns; Cr(VI) is a carcinogen
09 Final Rinse/Dry	Hot water/air; residual chemistry	Burns from hot dryer; slip hazard

 **HEADS-UP**
Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting plating chemicals — especially heavy metals like zinc, and chromium. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 6

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read. **Locate, before your first shift on the line:** the nearest **emergency eyewash** (within ~10 seconds' reach of every chemical station), the nearest **safety shower**, the **fire extinguisher** and alarm, the **rectifier emergency stop / disconnect** and main power shutoff, and the **spill kit** and **SDS binder** (Safety Data Sheets — the official hazard documents for every chemical on site).

**HEADS-UP**
Eyewash and shower stations must always be unblocked. Never stack parts, totes, or drums in front of them. The day you need one is the day you can't move boxes out of the way.

FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Chemical on skin	Get to the safety shower; flush at least 15 minutes . Remove contaminated clothing while flushing — don't stop early because "it feels better." Get medical attention; bring the SDS. Caustic especially: flush even longer; it keeps burning.
Chemical in eyes	Go to the eyewash; hold eyelids open and flush at least 15 minutes , rolling your eyes. Do not rub. Do not stop early. Get medical attention immediately. Bring the SDS.
Inhaled fumes/mist, feel ill	Move to fresh air immediately. Report it — do not "tough it out." Seek medical attention for anything beyond brief, mild irritation. Bring the SDS.
Chemical swallowed	Do not induce vomiting unless the SDS directs it (vomiting acid or caustic burns the throat again). Follow SDS first aid; call medical help / poison control immediately.
Electrical shock / arc-flash	Do not touch the person if they may still be in contact with current. Cut power at the emergency stop / disconnect first, then call for help and render aid.
Chemical spill	Alert others and keep people away. Only clean it if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. Never let spills reach floor drains.

**HEADS-UP — LOCKOUT/TAGOUT (LOTO)**
LOTO is mandatory for *all* rectifier and electrical maintenance. LOTO means physically locking the power off and tagging it so no one can switch it back on while someone is working on it. Never work on a rectifier that hasn't been locked out — "I'll just be a second" is how people get electrocuted.

**HEADS-UP — A CRITICAL ACID-MIXING RULE, FOR LIFE**
When diluting acid, **always add ACID to WATER — never water to acid**. Adding water to concentrated acid releases heat so violently it can boil and erupt the acid out of the container into your face. The memory aid: "**Do as you oughta — add acid to water.**" Pour slowly, down the side, acid going *into* a larger volume of water. **The same logic applies to dissolving caustic:** add caustic *slowly to water* with stirring — it gets very hot, and dumping it in fast can boil and spatter.

**FUN FACT**
The reason acid-to-water (and caustic-to-water) matters is heat capacity: a big pool of water absorbs the heat of dilution far better when the chemical is the small addition. Get it backwards and the tiny bit of water flash-boils and spits chemical. This one rule has saved more faces than any other in the chemical trades.

CHAPTER 7

THE SAFETY PHILOSOPHY OF AN ALKALINE ZINC LINE

THE MINDSET THAT KEEPS YOU SAFE ACROSS A WHOLE CAREER

Rules and PPE are the *tools*. The philosophy is the *mindset* that makes them work. Five principles guide everything on this line.

1. ASSUME EVERY TANK CAN HURT YOU.

Even the rinse tanks carry residual chemistry. Even “just water” can be slippery enough to put you on the floor. On *this* line, remember that the plating tank itself is a strong caustic — the “main event” tank is also a serious burn hazard. Respect comes first; familiarity is where complacency — and accidents — begin.

2. CONTAINMENT BEATS CLEANUP.

Move parts deliberately. Let them drain over the tank before transferring. Don't fling racks, splash, or rush handoffs. Most chemical exposures aren't dramatic spills — they're splashes and drips from sloppy handling. Slow, smooth movement is safer *and* makes better parts (less drag-out means cleaner rinses and a healthier bath).

3. RESPECT THE TWO FACES OF EVERY HAZARD.

The things that make good parts are the same things that can hurt you. The caustic that dissolves zinc into the bath burns your skin. The current that deposits zinc can arc-flash. The chromate that protects the part is toxic. You can't separate the work from the hazard — so you manage both at once, every time.

4. THE CHEMISTRY REWARDS DISCIPLINE.

Alkaline zinc is forgiving of iron but unforgiving of a drifting ratio, carbonate buildup, and sloppy additive control. The same discipline that keeps the bath healthy — careful rinsing, no shortcuts, controlling drag-in, watching the NaOH:Zn ratio — is the discipline that keeps *you* healthy. Good housekeeping is a safety practice, not just a quality practice.

5. WHEN IN DOUBT, STOP AND ASK.

No one expects you to have every answer, especially while learning. The unsafe operator is not the one who asks questions — it's the one who guesses. If a fume seems strong, a glove looks worn, a reading looks wrong, or a procedure feels off, *stop and check*. That instinct is the single most valuable safety habit you can build.

• THE THREE HAZARD FAMILIES TO INTERNALIZE

- Acids & Alkali (corrosives).** On this line the alkali is *everywhere*: the hot caustic cleaner (Station 1) **and** the hot caustic plating bath (Station 5). The mineral acids show up at activation (Station 3) and the bright dip (Station 7); the passivate is acidic too. *Defense*: goggles, face shield, gloves, apron — and the 15-minute flush rule if exposed. Never forget acid-to-water and caustic-to-water.
- Electrical & physical.** The rectifier runs high amperage — shock and arc-flash risk. Wet floors mean slips. *Defense*: LOTO for all electrical work, never touch a rectifier or bus bar with wet hands, non-slip boots, keep walkways clear and dry.
- Chromate / passivation chemistry.** Trivalent Cr(III) is an acidic irritant; hexavalent Cr(VI), if present anywhere, is a carcinogen with strict regulatory controls. *Defense*: know which your shop uses, wear the specified PPE, never let chromate mist go uncontrolled, follow OSHA requirements to the letter if any Cr(VI) is in play.



REMEMBER

Acids and alkali, electrical and physical, chromate chemistry. Three families. If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself.



FUN FACT

Many veteran platers can identify a bath problem by the look of a deposit before any instrument confirms it — a genuinely useful skill built over years. But notice the lesson hidden in that: *they got to be veterans by never getting hurt*. Skill and safety grow together. Build both, every shift, and you'll have a long, healthy career in this trade.



REMEMBER

You've now got the whole foundation: what plating is, what makes alkaline zinc special, how the nine-station line works as one connected system, why the order is locked by chemistry, what to wear, what to do in an emergency, and the safety mindset that ties it all together. Turn the page knowing the *why* — and the *how* will make a lot more sense.

PART TWO — THE PREP LINE (STATIONS 01-04)
STATION 01 OF 09

ALKALINE SOAK CLEAN

“PLATE OVER DIRT AND YOU’VE JUST PLATED DIRT.”



REMEMBER — THE PREP-LINE PROMISE

By the time a part reaches the zinc tank, 80% of whether it'll turn out beautiful or scrap has already been decided. The four prep stations — Clean, Rinse, Activate, Rinse — are not the “boring stuff before the real plating.” As the old-timers say: *“The plating bath gets the credit, but the pretreatment does the work.”*

Welcome to the very first tank. Your job here is simple to say and important to nail: **get the part perfectly clean**. Not “looks clean.” *Chemically* clean — no oil, no grease, no fingerprints, no shop dust. Do this right and you’ve made every job after you easier. Cut a corner here and it comes back as peeling, hazing, or spotty plating that nobody can fix later.

• WHAT YOU’RE DOING & WHY IT MATTERS

When steel parts arrive, they’re *not* clean — even if they look shiny. They almost always carry three layers:

1. **Particulate** — loose solid bits: shop dust, metal fines from machining, grit.
2. **Oil and grease** — the slippery film from stamping oils, cutting fluids, rust-preventives, and hand contact. Usually invisible, but there.
3. **Oxide scale** — a thin layer of rust/iron oxide bonded to the steel itself. (We don’t remove this here — that’s Station 3’s job. But it’s good to know it’s there.)

Soak Clean removes the first two. The cleaner is a hot, strongly alkaline, water-based solution. It works several ways at once: **heat** softens and thins the oils; **surfactants** (the soap-like ingredients) surround oil droplets and lift them off — like dish soap on a frying pan; and **builders / alkalinity** chemically break down (saponify) certain oils into something water can carry away.



TIP — THE WATER BREAK TEST (FREE, FAST, AND NEVER LIES)

A truly clean steel surface loves water. When you pull a clean part out, the water clings and **sheets off in one continuous, unbroken film**. A dirty surface repels water, so the water **beads up**. **Continuous sheet = PASS**. **Beading or broken film = FAIL** (clean it again). Use it on every rack, every time.

FAIL - water beads up

```
+-----+
| o o o | X
| o o |
+-----+
```

Oil remains -> RE-CLEAN

PASS - water sheets evenly

```
+-----+
|#####| OK
|#####|
+-----+
```

Surface clean -> proceed



TECHNICAL STUFF

“Water break” refers to the *surface energy* of the metal. Clean steel has high surface energy, so water spreads (low contact angle). Even a one-molecule-thick film of oil drops the surface energy, so water pulls itself into beads (high contact angle). That’s why the test catches contamination far too thin for your eye to see.

• OPERATOR ESSENTIALS

MEMORIZE THE RANGES

PARAMETER	RANGE	WHAT TO WATCH
Temperature	130–160°F (54–71°C)	Do not exceed 180°F
Time	3–10 min	Heavy oil: 8–10 min · Light stamping oil: 3–5 min
Concentration	4–8 oz/gal (30–60 g/L)	Titrate (test) weekly at minimum
pH	11.0–13.5	Depends on the cleaner formulation
Agitation	Air or mechanical, continuous	Still soak for tubular parts
Filtration	Surface skimmer recommended	Removes floating oil
Water Break Test	Pass = continuous sheet	The only reliable field test

A few plain-language definitions: **Concentration** = how much cleaner chemical is dissolved in the water (oz/gal = ounces per gallon). **Titrate** = a simple lab test that measures the actual concentration so you know when to add more. **Agitation** = movement in the tank that keeps fresh cleaner contacting the surface. **Surface skimmer** = a device that pulls floating oil off the top so it doesn't redeposit on your clean parts.

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HEADS-UP — SAFETY: HOT ALKALINE SOLUTION
This tank is **hot AND caustic**. Splashes cause **chemical burns on contact**, and the steam/mist irritates your airways. **PPE every time:** chemical splash goggles, face shield (when charging or dumping the tank), elbow-length chemical gloves, apron, chemical-resistant boots.
First aid: **skin contact** → **flush 15 minutes**; **eye contact** → **eyewash 15 minutes** and get help. Never lean over the tank to look in without eye protection — a stray bubble burst can reach your face.

• STEP-BY-STEP PROCEDURE

1. **Check the tank before you start.** Confirm temperature reads **130–160°F** (never above 180°F). Confirm concentration is in range (check the last titration log). Make sure the skimmer is running and the surface is reasonably free of floating oil.
2. **Inspect the load.** Heavy rust-preventive oil → plan for the long end of the time range (8–10 min). Light stamping oil → 3–5 min is plenty.
3. **Rack or load the parts** so solution reaches every surface. Avoid nesting parts where air can be trapped (trapped air = a dry spot = a dirty spot). For tubular/blind-hole parts, position them so the bore floods and drains.
4. **Immerse fully** and start your timer. Confirm agitation is working. For tubular parts, your shop may specify a still soak instead.
5. **Soak for the full time.** Don't rush it. The clock is doing chemistry you can't see.
6. **Withdraw slowly**, letting solution drain back into the tank (saves chemical, reduces drag-out).
7. **Run the water break test.** Look for a continuous sheet. **Pass** → **Station 2**. **Fail** → **back in the cleaner**.
8. **Log it** if your shop logs cleaning, and move the rack promptly so the clean surface doesn't sit and flash-rust.

• **TEACH-BACK CHECKLIST**

A trainee is signed off on Station 01 only when they can, unprompted:

- ☐ Name the two kinds of contamination the soak clean removes (particulate and oil/grease — it does NOT remove oxide scale; that’s Station 3).
- ☐ State the temperature range and the hard ceiling (**130–160°F; never over 180°F**).
- ☐ State the time for heavy oil vs. light oil (8–10 min vs. 3–5 min).
- ☐ Correctly perform a water-break test and read the result honestly (continuous sheet = pass).
- ☐ Verbally walk through what they’d do if the water-break test fails (re-clean it — never send it forward).
- ☐ Name three pieces of required PPE.

• **TOP MISTAKES NEW OPERATORS MAKE**

1. **Skipping the water break test because the part “looks fine.”** Oil that ruins plating is invisible. Test every time.
2. **Running the tank too cold.** Cold cleaner can’t cut oil no matter how long you soak. If the water break keeps failing, check temperature first.
3. **Letting the concentration drift low.** Cleaner gets used up. If nobody’s titrating, you’re slowly plating dirt without knowing it.
4. **Pulling parts through the floating oil layer.** If the skimmer is off or overwhelmed, the oil you just removed redeposits on the way out.
5. **Nesting parts so tight that air pockets form.** Wherever solution can’t reach, the part stays dirty. Rack with space.
6. **Going way too hot to “clean faster.”** Above 180°F you can flash off solution, etch or discolor parts, and create more mist hazard. Hotter is not better.

• **QUICK TROUBLESHOOTING**

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Water break fails	Low concentration or temp	Titrate cleaner; bring temp to range
Oily film after cleaning	Floating oil redepositing	Clean/run skimmer; dump tank if heavily contaminated
White residue on parts	Hard water salts; cleaner residue	Check rinse water quality
Etching/discoloration	Cleaner too aggressive or too hot	Reduce temp; consider milder formulation
Spotty adhesion downstream	Uneven cleaning; trapped air	Reposition on rack; add agitation

SIGN-OFF — STATION 01

Trainee: _____ Trainer: _____ Date: _____

“I confirm the parts in this load passed the water break test (continuous sheet, no beading), the soak clean was at 130–160°F and in concentration, and I wore required PPE.”

STATION 02 OF 09

RINSE (PRE-ACTIVATION)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

You just got the part beautifully clean — but it’s now coated in a thin film of the cleaner itself. This station’s whole job is to **wash that cleaner film off** before the part goes into the acid tank. It sounds like a “nothing” step. It isn’t: a weak rinse here quietly sabotages the acid tank next door, and you won’t see the damage until parts come out wrong.

• WHAT YOU’RE DOING & WHY IT MATTERS

The soak cleaner is **alkaline** (high pH). The very next tank — acid activation — is, well, **acid** (low pH). Acid and alkaline are chemical opposites: when they meet, they neutralize each other and cancel out. So if you carry a film of alkaline cleaner straight into the acid tank, three bad things happen:

- 1. **You waste acid** — the acid gets used up neutralizing the cleaner instead of doing its real job on the part.
- 2. **You raise the acid tank’s pH** — making it weaker and less effective at activating the steel.
- 3. **You reduce activation quality** — which shows up later as poor adhesion (plating that peels).

This invisible carry-over of liquid from one tank to the next is called **drag-out** (what leaves the tank clinging to the part) or **drag-in** (what arrives in the next tank). The rinse station’s job is to knock that drag-out down to almost nothing — by dilution. Clean water surrounds the part and the cleaner film dissolves off into the rinse water, where it’s diluted to harmlessness and flushed away by the constant flow of fresh water.

**TIP — HOW WE MEASURE A CLEAN RINSE: CONDUCTIVITY**

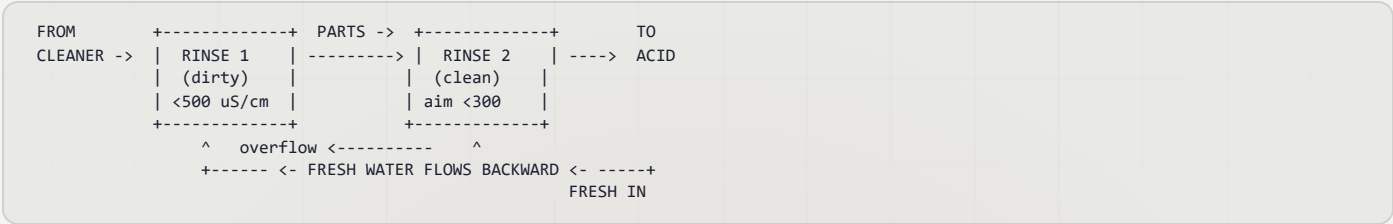
You can’t see dissolved cleaner, so we measure it with **conductivity** — how well the water conducts electricity. Pure water barely conducts at all. Dissolved chemicals make water conduct *more*. So **higher conductivity = more cleaner left in the rinse = dirtier rinse**. We measure it in **microsiemens per centimeter (µS/cm)** — just think of it as “the contamination number.” Lower is cleaner.

**REMEMBER — THE RULE CARD NUMBER: < 500 MS/CM**

Keep this rinse **below 500 µS/cm** (aim for **under 300**). Above 500, you’re dragging alkaline cleaner into the acid activation tank — wasting acid, raising its pH, and weakening activation. This number is your early-warning system. Check it daily.

• THE SMART WAY TO RINSE: COUNTER-FLOW

You *could* just blast fresh water through a single tank and dump it. It works, but wastes enormous water. The professional approach is a **counter-flow (counter-current) rinse**: the parts move forward; the water flows backward. Fresh water enters the *last* (cleanest) tank, then overflows backward into the dirtier tank. So your part gets progressively cleaner rinses, and the cleanest water it ever sees is the last thing to touch it before the acid tank.



• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F / 16–29°C)	No heating required
Time	30–60 sec with agitation	Longer for blind holes/tubing
Water Source	Municipal or DI	DI preferred
Conductivity	< 500 µS/cm (target < 300)	Monitor daily
Flow Rate	2–4 gal/min	Continuous overflow or counter-current
Tank Turnover	4–6 / hour	Prevents stagnation
pH	7–10	Elevated pH = cleaner drag-in

Plain-language definitions: **Ambient** = room temperature, no heater. **DI water** = deionized water with minerals removed (low conductivity, rinses better, no spots). **Continuous overflow** = fresh water in, dirty spills out the top. **Tank turnover** = how many times per hour the whole tank volume is replaced; 4–6 keeps it from going stale.

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HEADS-UP — SAFETY: RESIDUAL ALKALINE CLEANER
The rinse water carries dragged-in cleaner and starts out mildly caustic. Wear **goggles, chemical gloves, and non-slip footwear** — the floor around rinses is always wet, and slips are the #1 injury here.

• STEP-BY-STEP PROCEDURE

1. **Check conductivity at shift start.** If it's near or above 500 µS/cm, tell your lead — the rinse needs more flow or a refresh.
2. **Confirm water is flowing** (continuous overflow, or counter-current cascade active).
3. **Transfer the rack promptly** from the cleaner — dried cleaner is harder to rinse off.
4. **Immerse fully with agitation, 30–60 seconds.** For blind holes and tubing, go longer and move the part to flush trapped solution.
5. **Lift and drain** over the rinse tank for a few seconds to carry less forward.
6. **Move promptly to the acid tank** — don't let the rinsed part sit and flash-rust.

• TEACH-BACK CHECKLIST

- ☐ Explain why carrying cleaner into the acid tank causes problems (neutralizes acid — wastes acid, raises pH, weakens activation).
- ☐ Name the term for chemistry clinging to a part from the previous tank (drag-out / drag-in).
- ☐ Explain what conductivity tells us and which direction is “clean” (lower = cleaner).
- ☐ State the conductivity limit and target (< 500 µS/cm; target < 300).
- ☐ State how long to rinse and why longer for tubing (30–60 sec; tubing traps solution).

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
High conductivity	Heavy drag-out; low flow	More drain time off cleaner; more rinse flow; add counter-current stage
Soapy residue on parts	Inadequate rinse time	Extend dwell; add agitation
Acid tank going alkaline	Cleaner carry-over	Improve this rinse; add a second stage

SIGN-OFF — STATION 02
Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm
“I confirm the pre-activation rinse conductivity was within limit (< 500 µS/cm), water flow was running, and parts were rinsed the full time with PPE worn.”

STATION 03 OF 09

ACID ACTIVATION

“NO ACTIVATION, NO ADHESION. PERIOD.”

This is the station that makes plating *stick*. Your clean part still has an invisible, microscopic layer of oxide (think of it as super-thin rust) bonded to the steel. Zinc can’t grab onto oxide — it can only grab onto bare, chemically “awake” steel. The acid dip dissolves that oxide and leaves a fresh, reactive surface ready to bond with zinc. This is also the station with the **most safety weight** in your section — strong acid, fumes, and a hidden risk called hydrogen embrittlement.

• WHAT YOU’RE DOING & WHY IT MATTERS

Even a spotlessly cleaned steel part has a thin film of **oxide** on its surface. Oxide is what forms when iron meets oxygen — at its worst it’s visible rust and scale; at its mildest it’s an invisible film that forms in *minutes* on bare steel exposed to air. Either way, it’s a barrier. Zinc plated on top of oxide doesn’t truly bond — it sits on the oxide like paint on a greasy wall, and it will eventually flake or peel.

Acid activation (also called the **acid dip** or **pickle**) fixes this. The acid chemically dissolves the oxide layer right off the surface, leaving **clean, chemically active steel** — the bare iron atoms exposed and ready to form a strong metallurgical bond with the zinc. You’ll often see tiny bubbles during the dip — that’s **hydrogen gas (H₂)**, a normal sign the acid is reacting with the metal (and a clue to a hidden hazard below).



TECHNICAL STUFF

The reaction is straightforward: the acid (say HCl) reacts with iron oxide to dissolve it (iron oxide + acid → soluble iron salt + water), and once the oxide is gone the acid begins attacking the iron itself, producing hydrogen gas ($\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2\uparrow$). That last reaction is why you must *time the dip* — a little attack cleans and activates; too much etches the part and floods it with hydrogen.



REMEMBER — THE RULE CARD: 15–60 SECONDS

The dip time is everything. **Over-dip** = you over-etch the surface (roughness, pitting) AND drive excess hydrogen into the steel. **Under-dip** = oxide remains, and your plating won’t stick. Find the right time and *time it*.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient–110°F (Amb–43°C)	Ambient typical; warm only for heavy scale
Time	15–60 sec	15–30 for clean steel · 30–60 for cast iron/heavy oxide
Acid — Option A	HCl 10–50% v/v	Most common; produces fumes — ventilation required
Acid — Option B	H ₂ SO ₄ 5–25% v/v	Less fuming; slightly slower
Agitation	Manual rack movement only	No air agitation — it creates acid mist
Iron Content	Dump when > 50 g/L Fe	Spent acid loses bite; iron can drag forward
Inhibitor	Per vendor TDS (optional)	Reduces base-metal attack and hydrogen absorption

Plain-language definitions: **HCl** = hydrochloric acid (also called muriatic acid). **H₂SO₄** = sulfuric acid. Both are strong mineral acids. **% v/v** = “percent by volume” — how much concentrated acid is mixed into the water (50% v/v HCl means half acid, half water). **Inhibitor** = an additive that lets the acid keep dissolving *oxide* while slowing its attack on the *base steel itself* — protecting the part and cutting hydrogen absorption. **Iron content (Fe)** = dissolved iron that builds up as the acid works; over 50 g/L the acid is “spent.”

**HEADS-UP — SAFETY: STRONG MINERAL ACID (READ THIS TWICE)**

HCl fumes at room temperature — you don’t have to heat it to release irritating, corrosive vapor; ventilation is required. **H₂SO₄ causes severe dehydration burns** and reacts violently with water — mixing the two releases a burst of heat. (That’s exactly why you dilute by trickling a *small* amount of acid into a *large* volume of water: the big pool of water soaks up the heat so it can’t flash to steam and spatter. Pouring water onto acid does the reverse and is dangerous.) **Hydrogen gas (H₂) released during pickling is flammable** — no open flames or sparks near the tank.

PPE: goggles + face shield, acid-resistant gloves (butyl or neoprene), apron, boots, **respirator if ventilation is inadequate**. Skin contact: flush 15 min. Eye contact: eyewash 15 min. **No air agitation** in this tank — bubbling air through acid throws a fine acid mist into your breathing space. Move the rack by hand instead.

• THE BIG ONE: HYDROGEN EMBRITTLEMENT**HEADS-UP — SAFETY & QUALITY: HYDROGEN EMBRITTLEMENT**

When acid attacks steel, it releases hydrogen. Most of it bubbles away as gas — but some sneaks *into* the steel as single hydrogen atoms, wedging into the spaces between the iron atoms (the “lattice”). On ordinary mild steel this is usually harmless. But on **high-strength / hardened steel**, that trapped hydrogen makes the metal **brittle** — and a brittle part can **crack and fail suddenly under load**, sometimes hours or days later.

The acid pickle is the single greatest source of hydrogen absorption in the entire line. Uninhibited acid on high-strength steel (**≥ 40 HRC / ≥ 180 ksi UTS**) can load the lattice in *seconds*.

What you do about it: Use the **minimum effective acid concentration and dip time** — don’t over-dip “to be safe.” **Always use an inhibitor on substrates over 40 HRC.** If a job is flagged high-strength or embrittlement-sensitive, **stop and check with your lead** — and remember that hardened parts require a post-plate **bake** (handled at Station 06) to drive hydrogen back out. **HRC** = Rockwell C hardness; **ksi UTS** = thousands of psi of ultimate tensile strength — treat both on a traveler as a “slow down and check” flag.

**TECHNICAL STUFF — AN ALKALINE-ZINC NUANCE**

You might hear alkaline zinc called “lower-embrittlement” than acid zinc. Partly true: because alkaline zinc plates at *lower* cathode efficiency, a slightly more porous deposit forms early that lets some hydrogen escape, and the slower plate can allow more to diffuse out. **But do not treat alkaline zinc as embrittlement-free.** The pickle still loads hydrogen heavily, the bath still gasses hydrogen at the part, and the **ASTM B850 bake requirement on >40 HRC steel still applies in full.** When in doubt, bake.

• STEP-BY-STEP PROCEDURE

- 1. **Check the job for hardness flags first.** If the traveler says high-strength, hardened, ≥ 40 HRC, or ≥ 180 ksi — pause and confirm the right process and that inhibitor is in use. Embrittlement risk comes before everything else.
- 2. **Confirm ventilation is running** and your PPE (including face shield) is on.
- 3. **Verify acid strength and the dip time** for this part type: 15–30 sec for clean mild steel; 30–60 sec for cast iron or heavy oxide/scale.
- 4. **Immerse the rack fully** and start your timer. **Agitate by gently moving the rack by hand** — never air.
- 5. **Watch the surface.** Light bubbling is normal. The dark oxide should visibly lighten/disappear. If it's not clearing, the acid may be too dilute or spent — don't just leave a hard part in longer; check the bath.
- 6. **Pull at the timed mark.** Don't "round up." Over-dipping etches and embrittles.
- 7. **Drain briefly** and move **promptly** to the pre-plate rinse (Station 4) — activated steel re-oxidizes fast in air.
- 8. **Watch the iron level over time.** When the bath hits > 50 g/L Fe (or your shop's limit), flag it for dumping/recharge.

• TEACH-BACK CHECKLIST

- ☐ Explain what the acid dip removes and why it matters (the oxide layer — zinc can't bond to oxide, only to bare active steel).
- ☐ Explain what happens if you over-dip vs. under-dip (over: etching + embrittlement; under: oxide remains, plating peels).
- ☐ State the dip time for clean steel vs. heavy oxide (15–30 sec vs. 30–60 sec).
- ☐ Explain why there is NO air agitation in this tank (air creates a hazardous acid mist).
- ☐ Explain hydrogen embrittlement and which parts are at risk (≥ 40 HRC / ≥ 180 ksi) — and that alkaline zinc still needs the bake.
- ☐ Name the two things that reduce embrittlement risk here (minimum dip time/concentration, and inhibitor on hard parts).
- ☐ Recite the rule for diluting acid (always add acid to water, never water to acid).

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Dark oxide remains	Acid too dilute; time too short	Increase concentration; extend time (<i>check embrittlement risk first</i>)
Pitting / etching	Acid too strong, too long, or too warm	Reduce concentration; shorten dip; add inhibitor
Adhesion failure downstream	Incomplete activation; residual oxide	Re-check acid strength/time; check for alkaline drag-in from Station 2
Excessive HCl fuming	High concentration/temp; poor ventilation	Reduce to minimum effective; check exhaust
Dark smut on surface	High dissolved iron; copper contamination	Dump and recharge acid; investigate copper source
H-embrittlement failures	Over-long pickle; no inhibitor; hard steel	Shorten pickle; add inhibitor; review hardness; ensure relief bake

SIGN-OFF — STATION 03

Trainee: _____ Trainer: _____ Dip used: _____ sec · Inhibitor? Y / N

"I confirm I checked the job for hardness flags, ran the acid dip at the timed interval (15–60 sec) with manual agitation only, ventilation and PPE in place, and used inhibitor where required."

STATION 04 OF 09

RINSE (PRE-PLATE)

THE GUARD RINSE — THIS RINSE PROTECTS THE MOST PRODUCTIVE TANK ON THE LINE

You're at the last station before the zinc plating bath — the tank that actually makes the company money. Your job here is to **wash the acid (and the dissolved iron it carries) off the part** so it doesn't burden that expensive, carefully-balanced plating bath. Think of this rinse as the **guard at the door** of the most valuable room in the building.

• WHAT YOU'RE DOING & WHY IT MATTERS

Coming out of the acid dip, the part is wet with acid — and that acid now contains **dissolved iron** it picked up while pickling. If you carry that into the plating bath, you cause three problems:

1. **You neutralize free caustic.** The alkaline zinc bath relies on a tight NaOH:Zn ratio. Dragging acid in consumes free caustic, lowering the ratio — which hurts throwing power and efficiency and forces constant correction.
2. **You drag in unwanted anions.** Chloride or sulfate from the pickle accumulate in a bath that doesn't want them.
3. **You add iron.** This matters less than in acid zinc, but it still loads the filter with sludge and, in excess, dulls the deposit.



REMEMBER — IRON IS MORE FORGIVING HERE, BUT DON'T GET SLOPPY

In the *acid* zinc process, dragged-in iron stays dissolved forever and ruins the bath. **Alkaline zinc is kinder:** iron precipitates as iron hydroxide and the filter removes it. *But* — heavy acid drag-in still wastes caustic, upsets the ratio, and overloads the filter. Keep this rinse tight: it's still your primary defense, just against a slightly different set of villains.

CONTAMINANT	SOURCE	EFFECT ON THE ZINC BATH	PREVENTION
Free acid	HCl/H ₂ SO ₄ drag-out	Neutralizes free caustic; lowers NaOH:Zn ratio	Drain well; tighter rinse
Iron (Fe)	Spent pickle	Filter load; dulls deposit in excess (but is filterable)	Refresh pickle; double rinse
Chloride / sulfate	Pickle anions	Slow accumulation of unwanted salts	Tighter rinse; double stage



TIP — CONDUCTIVITY, TIGHTER THIS TIME: < 200 MS/CM

Same measurement as Station 2, but here the limit is **under 200 µS/cm, target under 100**. Tighter, because drag-in here directly burdens the plating bath. Check it daily and don't let it creep.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F / 16–29°C)	No heating required
Time	30–60 sec with agitation	Double rinse recommended
Water Source	DI preferred	Minimizes contamination to plating bath
Conductivity	< 200 µS/cm (target < 100)	Protects ratio and filter
pH	4.0–8.0	Below 4 = excessive acid drag-in
Flow Rate	Continuous overflow, 3–5 gal/min	Counter-current ideal
Drain Time	15–20 sec minimum	Prevents diluting the plating bath

Plain-language notes: **Double rinse** = two rinse tanks in series; the first knocks off most acid, the second polishes to the tight target. **Drain time** matters most here — it's the last drain before the money tank. **pH below 4** in the rinse warns you're dragging in too much acid.

**HEADS-UP — SAFETY: DILUTE ACID RESIDUE**

The water here carries dragged-in acid. Diluted but still irritating. Wear **goggles, chemical gloves, and non-slip footwear**. The floor is wet — watch your footing.

• THE 30-SECOND CLOCK: GET TO ZINC FAST

**REMEMBER**

Move into the plating bath within ~30 seconds of leaving the rinse. A freshly activated, rinsed surface starts to oxidize in air almost immediately — and an oxidized surface plates poorly, undoing Station 3's work. Keep the part wet. Speed matters at this handoff.

• STEP-BY-STEP PROCEDURE

1. **Check conductivity** at shift start. If it's above 200 µS/cm, the rinse needs more flow or a refresh before you plate through it.
2. **Confirm water flow** (continuous overflow at 3–5 gal/min; counter-current cascade running if equipped).
3. **Transfer promptly from the acid tank** — don't let the activated, acid-wet part sit and re-oxidize.
4. **Immerse fully with agitation, 30–60 seconds**. If you have a double rinse, give both tanks their time.
5. **Drain a full 15–20 seconds** over the rinse before moving on. This step is non-negotiable here.
6. **Move into the plating bath within 30 seconds** of leaving the rinse.

• TEACH-BACK CHECKLIST

- ☐ Name the three problems acid drag-in causes (neutralizes free caustic / lowers the ratio; drags in unwanted anions; adds iron load).
- ☐ Explain why iron is *less* of a permanent problem in alkaline zinc than acid zinc (alkaline precipitates and filters it out).
- ☐ State the conductivity limit here vs. Station 2 (< 200 µS/cm, target < 100 — tighter than Station 2's < 500).
- ☐ State how long to drain before moving on, and why (15–20 sec; less drag-in burdens the bath less).
- ☐ State how fast the part must reach the plating tank after rinsing (within ~30 seconds, before it re-oxidizes).

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Conductivity high	Heavy acid drag-out; low flow	Increase drain time from pickle; increase rinse flow
Plating bath ratio drifting	Acid drag-in neutralizing caustic	Tighten conductivity; add a double rinse
Filter loading fast with sludge	Iron dragging in from spent pickle	Dump/refresh the pickle; improve this rinse
Parts oxidizing before plating	Transfer too slow; part drying out	Speed up — enter plating within 30 sec

SIGN-OFF — STATION 04

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm

"I confirm the pre-plate rinse conductivity was within limit (< 200 µS/cm), parts were rinsed and drained the full time, and moved to plating within 30 seconds with PPE worn."

**REMEMBER — THE PREP-LINE PROMISE**

Four stations, one mission: hand the zinc bath a surface that is **clean, contaminant-free, oxide-free, and still wet**. Clean it (01), rinse the cleaner off (02), strip the oxide (03), rinse the acid and iron off (04), and run to the tank. **Clean → rinse → activate → rinse**. Do these four right, every rack — and even, ductile zinc will reward you.

PART THREE — THE MAIN LINE (STATIONS 05–09)
STATION 05 OF 09 • THE MAIN TANK

ALKALINE ZINC PLATING

WHERE BARE STEEL BECOMES EVENLY-PROTECTED STEEL

Take a breath. Up to now, every station — cleaning, rinsing, acid activation, rinsing again — has had one job: get the steel ready. None of those tanks put any zinc on the part. This tank is the payoff. **Station 05 is where the zinc coating is actually created.** Everything before it was preparation; everything after it is protection and finishing. If you only deeply understand one station on the whole line, make it this one.

The big idea in one sentence: **we use electricity to pull dissolved zinc out of a caustic liquid and stick it, atom by atom, evenly onto your steel part.** The cast of characters:

- **The cathode** — *your part*. The negatively-charged side. **Cathode = the part = negative = where zinc lands.**
- **The anode** — in modern alkaline zinc, usually **bare steel** plates or bars on the positive side. Unlike acid zinc, these **do not dissolve to feed zinc** — they mostly evolve oxygen. (Some shops still hang zinc anodes; know which yours uses.)
- **The electrolyte** — the bath liquid: water heavily loaded with **caustic soda (NaOH)** and dissolved zinc as **zincate**. The highway the zinc travels on.

**HEADS-UP — THIS BATH IS THE MOST DANGEROUS TANK ON THE LINE**

Unlike acid zinc’s mild bath, the alkaline zinc *plating* tank is a **strong, often warm caustic** — a serious burn and eye hazard all by itself, before you even add the rectifier. Full PPE (goggles + face shield, elbow gloves, apron, boots) every time, and treat any splash as an emergency 15-minute flush.

**FUN FACT**

Because steel anodes don’t supply zinc, the bath slowly *runs down* on metal as you plate — the opposite of acid zinc, where dissolving zinc anodes keep metal topped up automatically. Operators feed alkaline baths by dissolving zinc in a separate “dissolver” tank (zinc balls in caustic) and pumping that concentrate in, or by scheduled chemical additions.

WHY CYANIDE-FREE ALKALINE ZINC? (AND WHAT IT TRADES AWAY)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Excellent throwing power (1.5:1–2:1)	Plates evenly into recesses, blind holes, complex shapes
Ductile, low-stress deposits	Parts can be formed/crimped after plating without flaking
Uniform thickness distribution	Less “thin in the middle, thick on edges” than acid zinc
No cyanide (modern version)	Vastly safer and simpler than legacy cyanide zinc
Forgiving of iron	Iron precipitates and filters out — no permanent buildup

**REMEMBER**

The headline tradeoff is **speed**. Alkaline zinc cathode efficiency is only **~50–80%**, versus acid zinc’s 95–98%. A big share of your current makes hydrogen gas instead of zinc, so **alkaline zinc plates 2–3× slower**. That lower efficiency is not all bad, though — it’s *part of why the throwing power is so good* (edges that try to race ahead just waste current on gas, letting recesses catch up). You pay in time and electricity; you’re rewarded with evenness.

Other tradeoffs: a **hotter, stronger caustic** safety burden; **carbonate buildup** over time; and **active anode/ratio management** because steel anodes don’t self-feed.

• THE BATH RECIPE: WHAT'S IN THE TANK AND WHY

The cyanide-free alkaline zinc bath is a balanced recipe of four kinds of ingredients. Know what each one *does*, and you'll know what goes wrong when it drifts.

1. ZINC (THE METAL THAT DOES THE COATING)

Dissolved as **zincate**, Zn(OH)_4^{2-} . **Rack:** 8–15 g/L • **Barrel:** 6–12 g/L. It is the coating. Too low: low efficiency, thin/dull plating, hazy low-current areas. Too high: efficiency rises but throwing power *worsens* (you start losing the very advantage you came for) and the deposit can get rough/dark at high current.

2. SODIUM HYDROXIDE (THE DISSOLVER AND CONDUCTOR)

Caustic soda, NaOH . **Rack:** 100–150 g/L • **Barrel:** 90–130 g/L. It dissolves the zinc into zincate and carries the current. Too low: zinc won't stay dissolved, throwing power drops, anodes/parts misbehave. Too high: efficiency drops further (more gassing), more caustic mist, faster carbonate formation.

3. THE $\text{NaOH} : \text{Zn}$ RATIO (THE MASTER CONTROL)

This is the single most important number in the tank. **Target ~10:1 (range ~8:1 to 12:1).**



REMEMBER

Don't chase zinc and caustic as two separate numbers — chase the **ratio**. *Higher ratio (more caustic relative to zinc) → better throwing power but lower efficiency (slower). Lower ratio (more zinc) → faster but worse distribution and possible roughness.* When a job needs maximum even coverage, the bath runs leaner on zinc (higher ratio). When speed matters more, it runs a bit richer (lower ratio). Most general work sits near 10:1.

4. SODIUM CARBONATE (THE UNINVITED GUEST)

Na_2CO_3 isn't added — it *forms* as caustic absorbs CO_2 from the air and as organics break down. It slowly builds up, raising viscosity, reducing conductivity, and dulling deposits. **Treat/dump when it climbs above ~60–90 g/L** (per your shop's limit). Removed by chilling the bath to crystallize it out (carbonate is less soluble cold) and filtering, or by partial decant/replacement.

AND THE ADDITIVE SYSTEM: PURIFIER + CARRIER + BRIGHTENER

Where cyanide-free alkaline zinc lives or dies — these organics do the job cyanide used to do, dosed *per TDS* and replenished by **amp-hours**. (Full table on the next page.)



TIP — THE FOUR NUMBERS TO KEEP IN YOUR HEAD

Zinc 8–15, NaOH 100–150, ratio ~10:1, carbonate <60–90. Keep those four straight and you can sanity-check any analysis sheet at a glance. Notice there's **no pH number** here the way there was in acid zinc — the bath is so strongly caustic that “pH” is effectively off the top of the scale; you control it through the *caustic concentration and the ratio*, not a pH meter.

• THE ADDITIVE SYSTEM — THREE PLAYERS, ONE TEAM

ADDITIVE	PLAIN-LANGUAGE JOB	IF LOW	IF TOO MUCH
Purifier / conditioner	Sequesters tramp metals (Pb, Cu, Cd, Fe); keeps low-current areas plating clean	Dark, skip-plated, hazy low-current areas; blistering	Wasteful; can cause haze
Carrier (grain refiner)	Refines grain, supports throwing power and coverage	Dull, rough, poor low-current coverage	Stress, haze, possible blistering
Brightener	Adds luster and leveling	Dull / matte deposit	Brittle, stressed, peeling deposit; burning



REMEMBER

In alkaline zinc, **dull or dark low-current areas (recesses, the middle of the part)** usually point to **additive/purifier problems**, while in acid zinc trouble tends to show first at the **high-current** edges. Different baths, different “tell-tale” zones. Where the defect lives tells you where to look.



TECHNICAL STUFF — WHY DOSE BY AMP-HOURS?

Additives get *consumed* by the plating reaction. The more zinc you’ve plated (amp-hours = amps × hours), the more additive you’ve used. A tank running hard all day burns additive faster than one sitting idle. Replenishing per amp-hour keeps the chemistry steady regardless of how busy the shift was.

• YOUR BEST FRIEND: THE HULL CELL

Before you ever dump chemistry into a production tank on a hunch, you test your theory in a tiny tank first — the **Hull Cell**, a small wedge-shaped test tank. You plate one little panel under standard conditions — commonly **267 mL at 1–2 amps for ~5–10 minutes** for alkaline zinc (exact current/time per your TDS). Because the cell is wedge-shaped, the panel sees **high current at one end and low current at the other all at once** — an X-ray of your bath. Read it against your supplier’s reference panels.

WHAT THE PANEL LOOKS LIKE	WHAT IT’S TELLING YOU
Bright/uniform across the whole panel	Bath is in balance
Dull / dark at the low-current (thin) end	Low purifier or carrier; tramp-metal contamination
Dull / matte overall	Low brightener
Burnt / dark at the high-current end	Ratio too low (zinc too high), or excess brightener
Skip-plating / no deposit in low-current area	Strong metallic contamination (Pb, Cu) — needs purifier/treatment
Blistering	Organic breakdown or metallic contamination



TIP

Run a Hull Cell *before* and *after* any chemistry addition. Before, to diagnose. After, to confirm the fix worked — on a 267 mL panel instead of risking your whole production tank. Cheapest insurance on the line.

METALLIC CONTAMINATION: THE TRAMP-METAL PROBLEM

This bath punishes certain *tramp metals* in the low-current areas. The purifier's whole job is to keep them in check.

CONTAMINANT	ACTION LEVEL	WHAT IT DOES TO YOUR PARTS	WHERE IT SNEAKS IN FROM
Lead (Pb)	~2–5 ppm	Dark/skip low-current areas; blisters	Impure zinc feed; hardware
Copper (Cu)	~10–30 ppm	Dark spots, dull patches	Brass fittings; corroded bus bars
Cadmium (Cd)	~5–10 ppm	Dull, dark low-current areas	Contaminated feed/anodes
Iron (Fe)	More tolerant	Mild haze in excess; precipitates and is filterable	Pickle drag-in; steel in the bath



REMEMBER

Compare this to acid zinc, where copper's action level was a brutal **0.5 ppm** and **iron** was the unfilterable nightmare. Alkaline zinc *flips* that picture: it shrugs off iron (precipitates and filters out) and tolerates more copper, but it's specifically sensitive to **lead** in the low-current areas — which is exactly what the **purifier** is there to chelate. Different bath, different villains.



TECHNICAL STUFF — HOW TO CLEAN THE BATH

Purifier additions sequester tramp metals so they plate out harmlessly or stay locked up. **Dummying** (plating onto scrap “dummy” cathodes at low current) selectively pulls contaminating metals out. **Carbon treatment / filtration** removes organic junk and broken-down brightener. For carbonate, **chill-and-filter** crystallizes it out. Do bath treatments only under your chemist's direction.



HEADS-UP — CHROMIUM DRAG-BACK

Hexavalent chromium drifting *backward* from the chromate (Station 08) into the zinc bath causes blistering and bare/skip-plating, and won't dummy out easily. Removal is a **reduce-then-precipitate** step (reduce Cr^{6+} to Cr^{3+} , then it precipitates in the caustic and is filtered out) — under your chemist's direction. Good rinsing between plate and chromate is your prevention.

THE SUPPORTING EQUIPMENT (AND WHY EACH PIECE EXISTS)

EQUIPMENT	SETTING / SPEC	WHY IT'S THERE
Anodes	Steel (modern), or zinc where used	Steel = inert current path; zinc fed separately. If zinc anodes: 99.99% SHG
Zinc feed	Separate dissolver or additions	Steel anodes don't supply metal — you must
Anode bags	Polypropylene (if zinc anodes)	Catch fine sludge so it doesn't cause roughness
Agitation	Air + cathode-rod (rack); barrel 6–12 RPM	Fresh chemistry at the surface; prevents gas pockets
Filtration	Continuous, 1–2 turnovers/hr	Removes precipitated iron/particulate. “Turnover” = full tank-volume through filter
Temperature	70–95°F (21–35°C), target ~80°F	Too cold = dull, poor coverage; too hot = additive breakdown & carbonate. Often needs <i>cooling</i>
Current density	1–4 A/dm ² (10–40 ASF) rack; 0.3–1.0 A/dm ² barrel	How hard you're driving. Lower than acid zinc because efficiency is lower
Anode:cathode ratio	Watch it (steel anodes)	Too little steel anode area → high anode CD → excess oxidation of brightener



TECHNICAL STUFF

“Current density” is current spread over the part's surface area (A/dm² or ASF). It's *not* total amps — it's amps per unit area. With steel anodes, watch the **anode** current density too: too small an anode area concentrates current there and **oxidizes (burns up) your organic additives faster**, raising your additive consumption. This is a quirk you won't meet in acid zinc.

ALKALINE ZINC TANK CROSS-SECTION — TANK ANATOMY



STANDARD OPERATING PROCEDURE (STATION 05)

1. **Confirm the part is clean and active.** It should have come straight from the pre-plate rinse and entered plating **within ~30 seconds**. A part that sat and oxidized plates poorly — send it back if in doubt.
2. **Check bath readiness:** temperature 70–95°F (target ~80°F); NaOH:Zn ratio near 10:1; agitation running (air + cathode rod, or barrel 6–12 RPM); filtration running; steel anodes present and connected (or zinc anodes bagged and above the solution line).
3. **Verify your chemistry is current** — zinc, caustic, ratio, purifier, carrier, and brightener analyzed and replenished on schedule (per TDS / amp-hours); carbonate below limit.
4. **Set the rectifier** to the correct current for the part's surface area and target current density (remember: lower CD than acid zinc).
5. **Load the part / start the cycle.** Plating time runs **longer than acid zinc** — typically 10–60 minutes for the same thickness, because efficiency is lower.
6. **Monitor while running** — watch for unusual color, heavy gassing, or noise; confirm agitation and filtration stay on.
7. **At cycle end, remove and drain over the tank**, then transfer promptly to the post-plate recovery rinse (Station 06).
8. **Log it** — amp-hours run, additions made, temperature, ratio. This log is how you dose additive correctly tomorrow.



HEADS-UP — HYDROGEN EMBRITTLEMENT (READ THIS TWICE)

Even though alkaline zinc is sometimes called “lower embrittlement,” **hard steel above 40 HRC / ~180 ksi still requires a relief bake**. The pickle loads hydrogen heavily and the bath gasses hydrogen at the part. Such parts must be **baked** after plating (covered at Station 06). Flag embrittlement-sensitive parts; don't assume.

COMMON DEFECTS — WHAT YOU’LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Dull / dark in low-current areas (recesses, middle)	Low purifier/carrier; tramp metal (Pb/Cu/Cd)	Add purifier per TDS; analyze metals; dummy; Hull Cell
Dull / matte overall	Low brightener; temp too low	Add brightener per amp-hour; check temp
Burnt / dark at high-current edges	Ratio too low (zinc too high); excess brightener	Adjust ratio (more caustic / less zinc); Hull Cell
Thin, slow plating	Low zinc; ratio too high; low temp (low efficiency is normal)	Check zinc/ratio; confirm temp; extend time
Hazy / cloudy deposit	Carbonate high; organics; excess caustic	Analyze carbonate (chill-treat); carbon treat
Blistering / poor adhesion	Tramp metal; organic breakdown; poor pretreatment	Analyze metals; check prep; reduce brightener
Roughness	Particulate; iron sludge; poor filtration	Check/replace filter; improve filtration
Brittle, cracking deposit	Excess brightener; high stress	Reduce brightener; rebalance per Hull Cell

SAFETY — THIS STATION DEMANDS YOUR RESPECT



HEADS-UP — HOT STRONG CAUSTIC BATH & MIST

The bath is a **strong, warm caustic** and air agitation throws up a fine **caustic mist** you must not breathe — local exhaust ventilation is mandatory. Full PPE: splash goggles *and* face shield, elbow-length chemical-resistant gloves, apron, boots. Caustic on skin → flush 15+ minutes and report; it keeps burning even when it doesn’t sting at first.



HEADS-UP — THE RECTIFIER IS HIGH AMPERAGE

That power supply pushes a *lot* of current — shock and arc-flash risk. Before any maintenance, reaching into the tank, or bus-bar work — **Lock Out / Tag Out (LOTO)** the rectifier. No exceptions, ever, no matter how quick the job seems.



HEADS-UP — DISSOLVING CAUSTIC & FEEDING THE BATH

When making up or boosting the bath, **add caustic to water slowly with stirring** — it gets violently hot. Never add water to solid caustic. Dissolving zinc in caustic also releases heat and **hydrogen gas** — keep the dissolver vented and away from sparks.

TEACH-BACK & TOP MISTAKES — STATION 05

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part — and which charge?
- Why are the anodes usually *steel*, and how is zinc replenished?
- The four target numbers: zinc, NaOH, ratio, carbonate?
- What does the NaOH:Zn ratio control (throwing power vs. speed)?
- The three additives and what each does?
- Dull/dark in the *low-current* area — where do you look?
- Why does alkaline zinc throw so well but plate so slowly?
- Why does iron behave *better* here than in acid zinc, but lead worse?
- Why does hard steel still need a relief bake here?

TOP MISTAKES TO AVOID

- Chasing zinc and caustic separately instead of the *ratio*.
- Ignoring carbonate buildup until deposits go hazy.
- Forgetting steel anodes don’t feed zinc (bath runs lean).
- Running too hot (breaks down additives, makes carbonate faster).
- Letting the purifier lapse (dark low-current areas, blisters).
- Assuming “alkaline = no embrittlement” and skipping the bake.
- Plating a part that sat too long and oxidized.

SIGN-OFF — STATION 05

I confirm that I can independently: verify bath readiness (temp, ratio, agitation, filtration, anodes/feed); read an analysis sheet against target ranges and the NaOH:Zn ratio; run and interpret a Hull Cell panel; identify which additive is responsible for a given appearance defect; recognize the main tramp metals and the purifier’s role; and state the hydrogen-embrittlement bake requirement for hardened steel. I understand that chemistry adjustments are made only by authorized personnel.

Operator: _____ Date: _____ Trainer: _____ Date: _____

STATION 06 OF 09

POST-PLATE RECOVERY RINSE & THE BAKE DECISION

WASH THE CAUSTIC OFF, RECOVER YOUR CHEMISTRY — AND DECIDE WHETHER A PART GOES TO THE OVEN

Your part just came out of a caustic-loaded plating bath, dripping with caustic and dissolved zinc. The next tanks are the acid bright dip and the chromate. Carry that caustic forward and you neutralize the bright dip and contaminate the chromate. This rinse is the wall between the high-pH plating bath and the low-pH finishing tanks — and a smart shop also uses it to *recover* dragged-out chemistry. It's also the decision point for the embrittlement bake.

★

REMEMBER
A rinse is never “just a rinse.” Each one is a *checkpoint* that protects the next tank. This one protects the bright dip and chromate from caustic — and can hand free chemistry back to the plating bath.

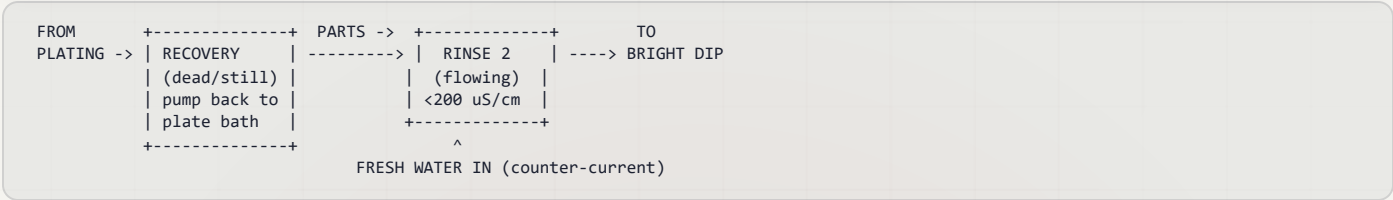
• WHAT AND WHY

A thin film of caustic zincate clings to every freshly-plated part. If that caustic rides into the acid bright dip, it **neutralizes the acid** (wasting it and weakening the brightening). If it reaches the chromate, it **raises the pH out of the critical window**, producing thin, blotchy, poorly-colored films and shortening bath life. The post-plate rinse washes that caustic film off *before* it reaches them.

TIP — THE DRAG-OUT RECOVERY (DEAD) RINSE
Many alkaline lines make the *first* post-plate rinse a still **drag-out recovery tank** (no flowing water, or very little). Its mildly-caustic, zinc-bearing water is periodically pumped back to top off the plating bath — recovering chemistry, cutting waste-treatment load, and reducing how much zinc and caustic you buy. A second flowing rinse then polishes the part clean. It's good chemistry *and* good economics.

• OPERATOR ESSENTIALS

PARAMETER	TARGET	WHY IT'S SET THERE
Conductivity (final rinse)	< 200 µS/cm (aim < 100)	Caustic drag-in is the whole concern
Temperature	Ambient	No heating needed
Time	30–60 sec with agitation	Recovery + flowing rinse strongly recommended
Drain time	15–20 sec over the tank	Excess water carried forward dilutes the bright dip/chromate
Water	DI preferred	Avoids hard-water haze on fresh zinc



• STANDARD OPERATING PROCEDURE

1. **Transfer promptly from plating.** Don't let the plated part sit and oxidize in air.
2. **Dip the recovery (dead) rinse first** if equipped, then **immerse the flowing rinse with agitation, 30–60 sec.** Make sure blind holes/recesses flush — trapped pockets hold caustic.
3. **Drain 15–20 seconds** over the tank before moving on.
4. **Handle with clean gloves only** — bare fingers leave oils/salts that show as fingerprints on bright zinc.
5. **Move promptly to the acid bright dip** (Station 07) before the zinc starts to oxidize/dull in air.

• RINSE QUALITY MONITORING (DAILY CHECKS)

CHECK	METHOD	HOW OFTEN	TAKE ACTION WHEN...
Conductivity	Meter	Every load	> 200 µS/cm → increase fresh-water flow
Visual clarity	Observe	Every load	Cloudy → dump/refill flowing rinse
Zinc content (final rinse)	Titration	Weekly	Climbing → increase overflow
Recovery tank level/strength	Per shop	Daily	Per schedule → pump back to plating bath

• THE HYDROGEN-EMBRITTLEMENT BAKE (CRITICAL FOR HARD STEEL)

This is the natural point in the line to talk about **baking**, because some specs require it *before* passivation. High-strength steel absorbs atomic hydrogen during pickling and plating, and that hydrogen can cause delayed cracking. **Baking** drives it back out of the steel before it can do damage.



HEADS-UP (SAFETY-CRITICAL — THIS PREVENTS CATASTROPHIC PART FAILURE)
For hardened steel **above 40 HRC**: bake **within 4 hours** of the last electroplating step, at **375 ± 25°F**, for a **minimum of 8 hours**, per **ASTM B850**. Some customer specs require the bake *before* passivation — check the spec. A missed bake on a structural fastener isn't a cosmetic defect; it's a part that can snap in service. **The fact that this is alkaline zinc does not exempt you** — bake when the spec or hardness calls for it.



TECHNICAL STUFF
“HRC” is the Rockwell C hardness scale. The higher the HRC, the harder — and the more brittle and embrittlement-prone — the steel. The 4-hour window matters because the longer hydrogen sits trapped in stressed steel, the more damage it can start before you bake it out.

• TROUBLESHOOTING (STATION 06)

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Bright dip neutralizing fast	Caustic drag-in from this rinse	Tighten conductivity; add/strengthen rinse stage
Chromate thin/blotchy	Residual caustic raising chromate pH	Improve rinse agitation and time
White haze on the zinc	Zinc oxidizing, or hard-water deposits	Use DI water; reduce transfer time
Plating bath running lean	Drag-out not being recovered	Pump recovery rinse back per schedule

• TOP MISTAKES & TEACH-BACK

TOP MISTAKES AT STATION 06

- Treating it as optional (it protects both finishing tanks *and* recovers chemistry).
- Letting conductivity climb above 200 µS/cm without more flow.
- No drain time — flooding the bright dip/chromate.
- Bare-handed handling of fresh zinc (fingerprints).
- Skipping or delaying the embrittlement bake on hard steel.
- Forgetting to pump the recovery tank back (wasted zinc/caustic).

TEACH-BACK — CAN YOU ANSWER?

- Why caustic drag-in matters at *this* rinse?
- The conductivity target, and what µS/cm measures?
- What a drag-out recovery (dead) rinse is, and why shops use it?
- The exact bake parameters (375 ±25°F, ≥8 hr, within 4 hr, >40 HRC, ASTM B850)?
- Why fresh zinc is handled with clean gloves?

SIGN-OFF — STATION 06

I confirm that I can independently: measure and interpret rinse conductivity; maintain the < 200 µS/cm target; operate the drag-out recovery rinse and pump-back; state and apply the ASTM B850 bake requirement for hardened steel (375 ± 25°F, minimum 8 hours, within 4 hours of plating); and handle fresh zinc without contaminating it.

Operator: _____ Date: _____ Trainer: _____ Date: _____

STATION 07 OF 09

ACID BRIGHT DIP / ACTIVATION

“ACTIVATION, ROUND TWO” — THIS TIME YOU’RE WAKING UP THE ZINC FOR THE CHROMATE

Here’s a station the acid-zinc line didn’t really need but alkaline zinc *does*. Zinc coming out of a strong caustic bath carries a faint residual caustic film and looks a little dull or smutty. Before the chromate, you give it a quick dip in **dilute mineral acid** that **neutralizes the leftover caustic, dissolves the dull film, and brightens the bare zinc** so the chromate forms an even, well-colored, well-adhered coating.



REMEMBER

Think of the bright dip as **activating the zinc for the chromate**, just as Station 03 activated the *steel* for plating. A weak, contaminated, or skipped bright dip is the #1 cause of dull, patchy, or poorly-colored chromate on an alkaline zinc line.

• WHAT YOU’RE DOING & WHY IT MATTERS

Two jobs in one quick dip:

1. **Neutralize & strip.** The dilute acid neutralizes the alkaline film still clinging from plating and lightly dissolves the outer smudge of dull/oxidized zinc, exposing fresh, reactive zinc.
2. **Brighten & uniform-prep.** A clean, uniformly-reactive zinc surface lets the chromate “bite” evenly everywhere — which is what gives you consistent color and full salt-spray performance.



TECHNICAL STUFF

The classic bright dip is **0.25–1% nitric acid (HNO₃)**, which both brightens and lightly passivates zinc. Some shops use **dilute sulfuric (0.5–2%)** to avoid nitric’s fumes and nitrate in the waste stream. Either way it’s *dilute and brief* — this is a flash dip measured in seconds, not a pickle. Over-dipping here strips zinc you worked hard to deposit.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Acid — Option A	Nitric (HNO ₃) 0.25–1% v/v	Best brightening; gives off irritating brown NO_x fumes — ventilate
Acid — Option B	Sulfuric (H ₂ SO ₄) 0.5–2% v/v	Less fuming; no nitrate in waste
Temperature	Ambient	No heating
Time	5–20 sec	Brief — a flash dip, not a pickle
Agitation	Gentle manual rack movement	No air (mist)
Strength control	Monitor; refresh when weak	Spent dip = dull, slow brightening



HEADS-UP — SAFETY: DILUTE MINERAL ACID (AND NITRIC FUMES)

Even dilute, this acid burns skin and eyes. If **nitric** is used, it gives off **brown, irritating nitrogen-oxide (NO_x) fumes** — ventilation is required and a respirator may be needed. **PPE:** goggles + face shield, acid-resistant gloves, apron, boots. Flush skin/eyes 15 min on contact. As always: **add acid to water, never water to acid** when making up the dip.

• STEP-BY-STEP PROCEDURE

- 1. **Confirm the part came cleanly from the post-plate rinse** with low caustic carry-over (heavy caustic neutralizes and spends the dip fast).
- 2. **Confirm ventilation** (especially with nitric) and PPE.
- 3. **Dip briefly — 5–20 seconds** — with gentle manual movement. Watch the zinc brighten.
- 4. **Pull promptly.** Don't over-dip — you're stripping zinc if you linger.
- 5. **Drain briefly**, then move **immediately** to the chromate (Station 08). A bright-dipped surface is highly reactive and must be chromated while fresh; many shops run the bright dip and chromate as a near-continuous pair, sometimes with only a quick rinse between (follow your shop's sequence — some skip the intermediate rinse to keep the surface active, others include a fast rinse to limit acid drag-in to the chromate).



TIP

Whether your shop rinses between bright dip and chromate depends on the chromate's tolerance for the bright-dip acid. Follow the posted sequence exactly — this is one spot where “improving” the process on your own can wreck color consistency.

• TEACH-BACK CHECKLIST

- ☐ State the two jobs of the bright dip (neutralize/strip the caustic film; brighten and uniform-prep the zinc for chromate).
- ☐ Explain why this is “activation round two” and what it activates (the zinc, for the chromate).
- ☐ State the acid type and strength (dilute nitric 0.25–1%, or dilute sulfuric) and the dip time (5–20 sec, brief).
- ☐ Explain what over-dipping does (strips deposited zinc).
- ☐ State the special fume hazard if nitric is used (brown NO_x fumes — ventilate).
- ☐ Explain why you must move to chromate promptly after the bright dip.

• TOP MISTAKES NEW OPERATORS MAKE

- 1. **Skipping or shortchanging the bright dip** → dull, patchy chromate later.
- 2. **Over-dipping** → stripping zinc and thinning the deposit.
- 3. **Running a spent dip** → weak brightening; refresh on schedule.
- 4. **Letting bright-dipped parts sit** before chromate → re-oxidized, uneven color.
- 5. **Ignoring nitric fumes** → ventilate and wear the right respirator.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Zinc still dull after dip	Dip too weak/spent; dip too short	Refresh/strengthen dip; extend slightly (still seconds)
Zinc etched/thin	Dip too strong or too long	Dilute the dip; shorten time
Patchy chromate downstream	Uneven bright dip; caustic carry-over	Improve post-plate rinse; refresh bright dip
Brown fumes over the tank	Nitric concentration high; poor exhaust	Check ventilation; verify dip strength

SIGN-OFF — STATION 07

Trainee: _____ Trainer: _____ Dip used: _____ sec · Acid: HNO₃ / H₂SO₄

“I confirm I dipped the parts briefly (5–20 sec) in the dilute bright-dip acid with ventilation and PPE in place, did not over-dip, and moved promptly to the chromate per the posted sequence.”

STATION 08 OF 09

PASSIVATE — TRIVALENT CHROMATE

THE CHROMATE CONVERSION COATING THAT TURNS BARE ZINC INTO REAL PROTECTION

You've made even, ductile zinc — surely that's the protection? The zinc *is* protective, but bare zinc is **reactive**. Left alone it quickly grows a powdery white corrosion product everyone calls **white rust** (zinc oxide/hydroxide). It looks terrible and shortens the coating's protective life. So we add one more coating *on top of* the zinc: a **chromate conversion coating**. As the tagline says: "*Zinc without passivate is a coating waiting to white rust.*"



REMEMBER

The zinc protects the steel. The passivate protects the zinc. It's protection on protection — and this is where that final shield goes on.

WHAT "CONVERSION COATING" MEANS (AND WHY IT'S DIFFERENT FROM PLATING)

Station 05 was **electroplating** — electricity *added* a separate layer of metal. Station 08 is a **conversion coating** — *no electricity*. You simply dip the zinc-plated part into the passivate, and the solution chemically *reacts with the surface of the zinc itself*, converting the very top layer into a thin, tough, protective film. The film isn't a separate thing laid on top; it's the zinc surface *transformed*. That's why it's called "conversion."

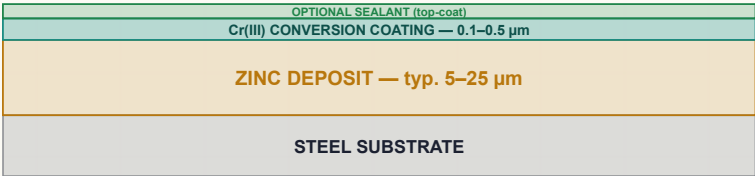


TECHNICAL STUFF

In the dip, the mild acid in the passivate lightly dissolves the outermost zinc, and chromium compounds deposit into that reacting surface, building a gel-like, chromium-rich film that dries hard. The result is a stable barrier that slows corrosion. (Older *hexavalent* Cr(VI) films could "self-heal" minor scratches by releasing soluble chromium that re-passivated bare zinc; modern *trivalent* Cr(III) films generally cannot self-heal — they protect purely as a stable barrier, which is partly why many specs add a topcoat/sealer.) The stack: steel substrate, zinc deposit (typically 5–25 µm), then the thin Cr(III) coating (0.1–0.5 µm), with an optional sealant on top.

PASSIVATE CROSS-SECTION (NOT TO SCALE)

CONVERSION = THE ZINC SURFACE TRANSFORMED



THE COLORS: CLEAR/BLUE VS YELLOW VS BLACK

Passivates come in colors. The color generally signals **how thick the film is** and **how much corrosion protection** you get — thicker chromate films are both more protective and more colored.

TYPE	LOOK	RELATIVE PROTECTION
Clear / Blue	Nearly clear, faint bluish	Good — entry-level protection
Yellow / Iridescent	Yellow with a rainbow shimmer	Highest of the common trivalent types
Black	Black	Decorative + protective



TIP

The customer's spec decides the color, not operator preference. Clear/blue for a clean bright look; yellow when maximum corrosion life is wanted; black for appearance. **Always match the spec.**

HEXAVALENT VS TRIVALENT: THE MOST IMPORTANT DISTINCTION ON THIS STATION

- **Trivalent chromium — Cr(III).** The modern, safer chemistry. **RoHS-compliant.** This is what the line is built around. Clear/blue, yellow, and black are all available as trivalent.
- **Hexavalent chromium — Cr(VI).** The legacy chemistry. It builds beautiful, durable, self-healing chromate films — but **it is a known human carcinogen.** It is *not* RoHS-compliant and is being phased out worldwide.

**HEADS-UP (SAFETY — THE SINGLE MOST IMPORTANT WARNING IN THIS MANUAL)**

Hexavalent chromium, Cr(VI), is a known carcinogen. The OSHA permissible exposure limit is just **5 µg/m³** of air. Where it is still used at all, it demands a **supplied-air respirator, a full protective suit, and medical surveillance per OSHA 29 CFR 1910.1026.** Its waste is **RCRA hazardous** and must be managed under your facility's permit. **Trivalent Cr(III) is the default for good reason — do not substitute hexavalent for trivalent without full regulatory and safety review and management sign-off.**

**FUN FACT**

That famous “rainbow shimmer” of an old-school yellow chromate? Classic Cr(VI) films were prized for it. Modern trivalent yellows now reproduce a similar look without the carcinogen — one of the quiet success stories of green chemistry in metal finishing.

• OPERATOR ESSENTIALS (EXACT NUMBERS BY TYPE)

The single most important control variable here is **pH** — every passivate has a narrow “critical pH window,” and outside it the film comes out too thin, too thick, non-adherent, or the wrong color.

PARAMETER	TRI CLEAR/BLUE	TRI YELLOW	TRI BLACK	HEX YELLOW (LEGACY)
Temperature	70–85°F	70–110°F	65–85°F	70–95°F
pH	1.5–2.2	1.8–3.5	2.0–3.5	1.5–2.0
Time	30–60 sec	45–90 sec	30–60 sec	10–30 sec
Cr type	Cr(III) RoHS	Cr(III) RoHS	Cr(III) RoHS	Cr(VI) NOT RoHS
Salt spray (white rust)	72–120 hrs	120–200+ hrs	72–120 hrs	96–240 hrs

**REMEMBER**

The rule-card number is “**pH per TDS.**” Why not a single number? Because the exact window depends on the specific product. The typical trivalent clear/blue window is **1.5–2.2**, but *your* TDS is the authority. pH is *the* control knob here — check it often.

**TECHNICAL STUFF — WHAT IS “SALT SPRAY”?**

It's an accelerated corrosion test (ASTM B117): parts sit in a sealed cabinet sprayed with salt fog, and you record how many *hours* until white rust appears. More hours = better protection. It's how the “72–120 hrs” numbers above are generated and how customers specify corrosion requirements.

**TIP (ALKALINE-ZINC-SPECIFIC)**

Alkaline zinc's even, ductile, lower-stress deposit generally **passivates very uniformly** and takes color well — often *more* evenly than acid zinc. **But** use a passivate **formulated for alkaline zinc.** Acid-zinc and alkaline-zinc passivates are tuned to different deposit surfaces and are *not* interchangeable — using the wrong one is a common cause of color and adhesion problems on this line.

• STANDARD OPERATING PROCEDURE (STATION 08)

1. **Confirm the part came cleanly through the post-plate rinse and the acid bright dip** (Stations 06–07) — low caustic carry-over, freshly brightened zinc. A dull or caustic-filmed surface chromates unevenly.
2. **Confirm any required embrittlement bake was done** if the spec calls for baking before passivation.
3. **Check the passivate bath:** correct type for the spec, **pH in the product's window** (typically 1.5–2.2 for tri clear/blue — verify against TDS), and temperature in range.
4. **Immerse for the specified time** — e.g., 30–60 sec for tri clear/blue; 45–90 sec for tri yellow. Time and pH together control film thickness and color.
5. **Drain over the tank**, then proceed to the final rinse and dry — **Stage 09**, on its own page next.

• TROUBLESHOOTING (STATION 08)

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Rainbow / iridescent (when it should be clear)	pH too low; immersion too long	Check pH; reduce time
Pale / thin film	pH too high; bath depleted	Lower pH; replenish per TDS
Blotchy / uneven color	Caustic carry-over; weak bright dip; uneven zinc	Improve post-plate rinse & bright dip; check plating
Film rubs off	pH out of range; insufficient cure	Check pH; allow the full 24-hr cure before handling
Passivate bath life short	Caustic/zinc drag-in; chloride from drag chain	Improve post-plate rinse (Station 06)



TIP

Notice how many passivate problems trace back *upstream* to the post-plate rinse and the bright dip. When this station misbehaves, look backward as well as at the passivate itself.

• INTEGRATED SAFETY (STATION 08)



HEADS-UP (TRIVALENT)

Even the “safe” trivalent passivate is **acidic (pH 1.5–3.5)** and **causes chemical burns**. Wear goggles plus face shield, chemical-resistant gloves, and an apron. Flush skin or eyes with water for **15 minutes** on contact.



HEADS-UP (HEXAVALENT — WHERE STILL USED)

Known carcinogen. OSHA PEL 5 µg/m³. Supplied-air respirator, full protective suit, and medical surveillance per **OSHA 29 CFR 1910.1026**. Waste is **RCRA hazardous** — manage per facility permit. The strong recommendation is to run trivalent.

• TEACH-BACK & TOP MISTAKES — STATION 08

TEACH-BACK — CAN YOU ANSWER?

- Why does plated zinc need a passivate — what is “white rust”?
- How is a conversion coating different from electroplating?
- What do the colors tell you about thickness/protection?
- Cr(III) vs Cr(VI) — key safety/regulatory facts for each?
- Why is pH the critical control variable; tri clear/blue window?
- Why use a passivate formulated *for alkaline zinc*?
- The two ways to wreck a good film during dry & cure?

TOP MISTAKES TO AVOID

- Letting pH drift out of the window (wrong color/thin film).
- Carrying caustic in from a weak post-plate rinse, or a weak bright dip.
- Using an acid-zinc passivate on alkaline zinc.
- Drying above 150°F (cracks trivalent films).
- Handling or testing before the 24-hour cure.
- Confusing trivalent and hexavalent.

SIGN-OFF — STATION 08

I confirm that I can independently: distinguish trivalent from hexavalent chromium and state the hazard and PPE for each; select and verify the correct passivate type and color for the spec; measure and hold pH within the product's critical window; run the immersion, final rinse, dry (≤150°F), and 24-hour cure correctly; diagnose common film defects and trace them upstream when needed; and apply the correct PPE.

Operator: _____ Date: _____ Trainer: _____ Date: _____

STAGE 09 OF 09 • THE FINISH LINE

FINAL RINSE & DRY

THE LAST 60 SECONDS — WHERE A PERFECT PART IS FINISHED, OR QUIETLY RUINED

The chromate film is on. You're one short stage from a finished part. Stage 09 has two simple jobs: **rinse off the leftover passivate chemistry** so it doesn't leave residue or keep reacting, and **dry the part gently** so the soft, fresh conversion film sets without cracking. It feels like the easy part — and that's exactly why parts get wrecked here. Trivalent conversion films are gel-like as they form and **crack if dried too hot** — cracks open paths for corrosion, defeating the whole point of passivating.



REMEMBER

The passivate film is **soft when fresh**. Heat cracks it; touching it rubs it off. Patience at Stage 09 protects every step upstream.



TIP — OPTIONAL SEAL/TOPCOAT

Many alkaline zinc jobs add an **optional seal or topcoat** after the final rinse, before drying, to boost salt-spray life, add lubricity, or set fastener torque-tension. Because trivalent films don't self-heal the way old hexavalent films did, a sealer is a common way to reach high salt-spray numbers. If your spec calls for a seal, apply it per its TDS *before* the final cure — it's an add-on, not a substitute for the chromate.

• OPERATOR ESSENTIALS

STEP	SETTING	WHY IT'S SET THERE
Final rinse	DI water, ambient, 15–30 sec, gentle immersion	Removes passivate residue without stripping the fresh film
Optional seal/topcoat	Per spec TDS, applied before dry	Boosts salt-spray / adds lubricity (if specified)
Dry temperature	120–150°F max (trivalent)	Film cracks above 150°F — do not over-dry
Cure time	24 hours minimum before packing/testing	The soft film needs time to harden fully
Handling	Clean gloves only	Bare hands mar the soft film and leave salts/oils



HEADS-UP (DON'T RUIN A GOOD FILM AT THE FINISH LINE)

Two easy ways to destroy a perfect passivate in the last 60 seconds: **drying too hot** (above 150°F cracks trivalent films) and **handling or testing before the 24-hour cure** (the soft film rubs off).

• STANDARD OPERATING PROCEDURE (STAGE 09)

1. **Final DI rinse:** gentle immersion in ambient DI water for **15–30 seconds**.
2. **Apply seal/topcoat if the spec requires it** (per its TDS), then drain.
3. **Dry gently** in warm air, **120–150°F max** — never hotter, or the trivalent film cracks.
4. **Set aside to cure** — at least **24 hours** before packing, salt-spray testing, or rough handling.
5. **Handle with clean gloves** throughout, and inspect for spotting, haze, or the correct color per spec.

• TEACH-BACK CHECKLIST & TOP MISTAKES

TEACH-BACK — CAN YOU ANSWER?

- The two jobs of Stage 09 (rinse residue; gently dry so the film sets).
- Max dry temperature and what happens above it (150°F; film cracks).
- Final-rinse water, temperature, and time (DI, ambient, 15–30 sec).
- Cure time before packing/testing (24 hr; film is soft); clean-glove handling.

TOP MISTAKES TO AVOID

- Drying too hot to “speed things up” (above 150°F cracks the film).
- Packing/testing before the 24-hour cure (soft film rubs off; false salt-spray).
- Skipping the final DI rinse (residue spots or keeps reacting).
- Bare-handed handling; skipping a required seal/topcoat.

SIGN-OFF — STAGE 09

Trainee: _____ Trainer: _____ Date: _____

“I confirm the part received a final DI rinse (and any required seal), was dried at 150°F or below, handled with clean gloves, and set aside for the full 24-hour cure before packing or testing.”

PART FOUR — BACK MATTER & CERTIFICATION

THE BIG-PICTURE TOPICS

ACROSS-THE-LINE REFERENCE

EMBRITTLEMENT & BAKING, RINSING & WATER, WASTE, AND QUALITY — GATHERED IN ONE PLACE

• HYDROGEN EMBRITTLEMENT & BAKING (CONSOLIDATED)

- **Who’s at risk:** steel ≥ 40 HRC / ≥ 180 ksi UTS (high-strength fasteners, springs, hardened structural parts).
- **Where hydrogen comes in:** mostly the **acid pickle** (Station 03); also the plating bath gassing hydrogen at the part (Station 05).
- **Defenses during processing:** minimum pickle time/concentration, **inhibited acid** on hard parts, prompt processing.
- **The relief bake (ASTM B850):** $375 \pm 25^{\circ}\text{F}$, **minimum 8 hours, started within 4 hours** of the last plating step. Some specs require it *before* chromate. Alkaline zinc is *somewhat* lower-risk than acid zinc, **but the bake requirement still applies in full** — never skip it on spec’d parts.

★

REMEMBER

“Alkaline zinc is lower-embrittlement” is a *relative* statement, not a free pass. When the hardness or the customer spec calls for a bake, you bake. A snapped structural bolt doesn’t care which bath plated it.

• RINSING & WATER

- **Why rinses matter:** each rinse is a *firewall* between chemistries (alkaline → acid → caustic plate → acid → chromate). Skip or weaken one and the next tank suffers.
- **Conductivity targets on this line:** pre-activation rinse $< 500\ \mu\text{S}/\text{cm}$ (target < 300); pre-plate guard rinse and post-plate rinse $< 200\ \mu\text{S}/\text{cm}$ (target < 100).
- **Water quality:** **DI water preferred** near the plating and chromate tanks (low conductivity, no hard-water spotting). **Counter-current & recovery rinsing** save water and recover chemistry; the post-plate **drag-out recovery (dead) rinse** is a signature alkaline-zinc economy — its caustic, zinc-bearing water pumps back to the plating bath.

• WASTE & ENVIRONMENTAL BASICS

- **No cyanide (modern bath)** — a major advantage of cyanide-free alkaline zinc; no cyanide-destruct step needed (legacy cyanide baths required alkaline chlorination).
- **Caustic & zinc** in rinse/drag-out: neutralize and precipitate zinc as hydroxide; recover via the drag-out rinse where possible. **Carbonate** is removed from the bath by chill-crystallization/filtration or partial replacement, not flushed down a drain.
- **Acids (pickle, bright dip):** spent acid carries dissolved iron/zinc — neutralize and treat per permit; **nitric bright dips add nitrate** to the waste stream (a treatment/permit consideration — one reason some shops choose sulfuric).
- **Chromate: trivalent Cr(III)** waste is far less hazardous than **hexavalent Cr(VI)** (RCRA hazardous; must be reduced to Cr(III) before precipitation). Always per your facility’s wastewater permit and SDS.
- **Golden rule:** **never let any process solution reach a floor drain.** All drag-out and spills go to treatment.

• QUALITY CHECKS (THICKNESS, ADHESION, CHROMATE, SALT SPRAY)

CHECK	HOW IT’S DONE	WHAT “GOOD” LOOKS LIKE
Thickness	Magnetic gauge, XRF, or cross-section/coulometric	Meets the SC minimum per ASTM B633 (5–25 μm). <i>Alkaline zinc’s edge: very even readings</i>
Adhesion	Bend/impact test; tape test	No flaking or peeling — ductile deposit usually passes easily
Chromate	Visual color/uniformity; film weight; Cr(VI) spot test	Even color per spec; uniform, adherent film
Salt spray	ASTM B117 cabinet, hours to white rust	Meets spec hours (e.g., tri yellow 120–200+ hr); test only <i>after</i> 24-hr cure
Water break	Free field test after cleaning	Continuous sheet = clean
Hull cell	267 mL test panel	Uniform/bright per reference; diagnoses bath drift

TIP

Alkaline zinc’s calling card on a thickness report is **uniformity**: low spread between thick (edge) and thin (recess) readings. If a customer’s main complaint about an acid-zinc job was “thin in the recesses,” moving to alkaline zinc often fixes it — at the cost of slower plating.

— YOUR DECODER RING

GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

**TIP**

Keep this glossary tabbed. The fastest way to sound like a veteran on the shop floor is to use the right word for the right thing.

Acid Activation (acid dip, pickle): Station 3. A dip in strong acid (HCl or H_2SO_4) that dissolves the invisible oxide on steel and “wakes up” the bare metal so zinc can grip it. *No activation, no adhesion.*

Acid Bright Dip: Station 7. A brief dip in *dilute* mineral acid (often 0.25–1% nitric) that neutralizes leftover caustic, brightens the zinc, and preps it for chromate. “Activation, round two.”

Adhesion: How well the plated zinc sticks to the steel. Good adhesion means it won’t peel or flake. Alkaline zinc’s ductile deposit adheres well.

Agitation: Movement of solution or parts. Keeps fresh chemistry against the part, knocks off hydrogen bubbles (which cause pits), evens the deposit. Air, mechanical, or barrel rotation.

Alkaline: The opposite of acidic; high pH. Both the soak cleaner *and* the alkaline zinc plating bath are alkaline (caustic).

Alkaline Zinc (cyanide-free / zincate): The process this manual is about. Zinc dissolved in **sodium hydroxide** as **zincate**, no cyanide. Excellent throwing power, ductile deposits, but slower (lower efficiency).

Ambient: Room temperature, roughly 60–85°F. No heating needed.

Amp-Hour (A-h): “How much plating you’ve done”: amps × hours. Additives are consumed per amp-hour, so you replenish based on amp-hours run, not clock time.

Anode: The positively-charged side. In modern alkaline zinc the anodes are usually **bare steel** (inert — they don’t supply zinc); zinc is fed separately. Some shops still use zinc anodes.

ASF: Amps per Square Foot — a unit of current density. Alkaline zinc rack work runs about 10–40 ASF (lower than acid zinc, because efficiency is lower).

ASTM B633: The common spec for zinc plating on steel. Sets minimum thickness by environment (Service Condition) and a **Type** for the supplementary finish — Type I = as-plated, Type II = colored chromate, Type III = clear/colorless chromate, Type IV = phosphate. The Type describes the chromate, *not* whether the bath is acid or alkaline.

ASTM B850: The spec covering the hydrogen-embrittlement relief bake.

Barrel Plating: Plating many small parts together by tumbling them in a rotating perforated barrel. *Alkaline zinc is especially good for barrel work* thanks to its throwing power.

Brightener: The organic additive that adds luster/leveling to the deposit. Too little = dull; too much = brittle/stressed.

Carbonate (Sodium Carbonate, Na_2CO_3): A contaminant that *builds up* in caustic baths as they absorb CO_2 and organics break down. Raises viscosity, dulls deposits. Removed by chill-crystallization/filtration. Treat/dump above ~60–90 g/L.

Carbon Treatment: Circulating the bath through activated carbon to remove organic contamination and broken-down additive.

Carrier (Conditioner / Grain Refiner): The additive that refines grain and supports coverage/throwing power. Too little = rough/dull, poor low-current coverage.

Cathode: The part being plated — the negatively-charged work. Zinc deposits onto the cathode.

Cathode Efficiency: The percentage of current that actually deposits zinc (versus making hydrogen gas). Alkaline zinc is only ~50–80% — the price of its great throwing power (acid zinc is 95–98%).

Caustic / Caustic Soda: Common names for **sodium hydroxide (NaOH)**, the strong base that is the heart of the alkaline zinc bath. Burns skin and eyes; handle with full PPE.

Chromate / Conversion Coating: See Passivation. A thin protective film grown chemically (not electroplated) on the zinc surface.

Conductivity ($\mu\text{S}/\text{cm}$): A measure of dissolved salts in rinse water — how “dirty” the rinse is. **Lower is cleaner.** Rinse targets tighten closer to the plating bath.

Counter-Current Rinse: A water-saving setup where fresh water flows opposite to the parts, so the cleanest water meets the cleanest parts.

Cr(III) / Trivalent Chromium: The modern, RoHS-compliant chromium used in today’s passivates. Much safer than hexavalent.

Cr(VI) / Hexavalent Chromium: The old form. A **known carcinogen** and NOT RoHS-compliant. Legacy use only, under strict OSHA controls.

Current Density (CD): How much current flows per unit of part surface area (ASF or A/dm^2). High-CD areas (edges, tips) and low-CD areas (recesses) behave differently. *Watch anode CD too with steel anodes — too high burns up additives.*

Cyanide Zinc: The legacy alkaline zinc that used sodium cyanide. Excellent plating but acutely toxic; replaced by cyanide-free zincate baths.

DI Water (Deionized Water): Water with dissolved minerals removed. Cleaner than tap; preferred near the plating and passivate tanks.

Drag-Out Recovery (Dead) Rinse: A still post-plate rinse whose caustic, zinc-bearing water is pumped back to the plating bath — recovers chemistry and cuts waste.

Drag-In / Drag-Out: Chemistry that rides along on the parts from one tank to the next. *Drag-out* leaves a tank; *drag-in* arrives in the next. Rinses control it.

Dummy / Dummying: Plating onto scrap “dummy” cathodes at low current to draw metallic contamination out of the bath.

Ductility: The ability of the deposit to bend/form without cracking. Alkaline zinc deposits are notably ductile — good for parts formed after plating.

Filtration: Continuously pumping the bath through a filter to remove particles and *precipitated iron*. Alkaline zinc runs roughly 1–2 turnovers/hr.

• GLOSSARY (CONTINUED)

Hexavalent: See Cr(VI).

Hull Cell: A small, trapezoid-shaped test tank that plates a sample panel across a *range* of current densities at once. Reading the panel tells you what's wrong with the bath.

Hydrogen Embrittlement (HE): When atomic hydrogen soaks into steel during pickling/plating and makes hardened steel **brittle** — it can crack later under load. The acid pickle is the single greatest source.

HE Relief Bake: Baking plated parts to drive hydrogen back out. For steel **>40 HRC**: bake **within 4 hours** of plating at **375 ±25°F for at least 8 hours** (ASTM B850). *Required for alkaline zinc too, despite its lower relative risk.*

HRC (Rockwell C Hardness): A hardness scale. Above **40 HRC**, steel is "high-strength" and at real risk of hydrogen embrittlement.

Inhibitor: An additive in the acid pickle that slows attack on the base metal and reduces hydrogen absorption. Required for >40 HRC steel.

Iron (Fe) Contamination: Dissolved iron from the pickle. In alkaline zinc it **precipitates as iron hydroxide and the filter removes it** — far more forgiving than acid zinc, where iron accumulates.

Leveling: The deposit's ability to fill tiny scratches for a smoother surface.

LOTO (Lockout/Tagout): De-energizing and locking power off before working on the rectifier/electrical equipment.

µm (Micrometer / Micron): A thickness unit. One micron = 0.001 mm ≈ 0.04 mil. Zinc deposits typically 5–25 µm.

Mil: One-thousandth of an inch (0.001"). 1 mil ≈ 25 µm.

NaOH:Zn Ratio: The master control number of the alkaline zinc bath. Target ~10:1 (range 8:1–12:1). *Higher = better throwing power but slower; lower = faster but worse distribution.*

Nitric Acid (HNO₃): The classic bright-dip acid for zinc (dilute, 0.25–1%). Brightens and lightly passivates; gives off brown NO_x fumes — ventilate.

Oxide: The thin film (rust/tarnish) on bare steel in air. Blocks adhesion, so acid activation dissolves it right before plating.

Passivation (Passivate / Chromate): Station 8. A trivalent-chromium conversion coating grown on the bare zinc that dramatically slows **white rust** and can add color. *Zinc without passivate is a coating waiting to white rust.*

pH: A 0–14 scale of acidity. The alkaline zinc plating bath is *off the top of the practical scale* (very high caustic) — you control it via caustic concentration and the ratio, not a pH meter. Trivalent clear/blue passivate sits at roughly 1.5–2.2.

Pitting: Tiny holes from hydrogen bubbles clinging to the part (low carrier/poor agitation) or from dirty parts.

ppm (Parts Per Million): A unit for very small concentrations, used for tramp-metal limits.

Purifier: The alkaline-zinc additive that sequesters tramp metals (lead, copper, cadmium, iron) to keep low-current areas plating clean. Lapse it and you get dark/skip low-current areas and blisters.

Rack Plating: Plating parts individually hung on a rack/fixture. Used for larger or cosmetic parts.

Rectifier: The power supply that converts AC to the DC current that drives plating. **High amperage — shock and arc-flash hazard.** Always LOTO before servicing.

RoHS: "Restriction of Hazardous Substances," a regulation banning certain hazardous materials (including hexavalent chromium) in many products. Trivalent passivates are RoHS-compliant; hexavalent is not.

Salt Spray Test: A corrosion test (ASTM B117) that sprays parts with salt fog and measures hours to white rust. Verifies passivate performance.

Seal / Topcoat: An optional final layer applied after the chromate to boost salt-spray life, add lubricity, or set torque-tension. Common with trivalent (which doesn't self-heal).

Service Condition (SC): ASTM B633's rating of environment harshness, setting minimum zinc thickness: SC1 (indoor mild, 5 µm), SC2 (indoor moderate, 8 µm), SC3 (outdoor moderate, 12 µm), SC4 (severe/marine, 25 µm).

SHG (Special High Grade) Zinc: 99.99%-pure zinc, used for the zinc feed/dissolver (and zinc anodes where used). Purity matters because impurities like lead contaminate the bath.

Skimmer: A device that removes floating oil off the soak-clean tank.

Sodium Hydroxide (NaOH): See Caustic.

Steel Anode: The modern alkaline zinc anode — inert; carries current and evolves oxygen but does **not** supply zinc (which is fed separately).

Throwing Power: The bath's ability to plate evenly into recesses. Alkaline zinc's throwing power is **excellent** (surface-to-recess ratio ~1.5:1–2:1, closer to 1:1 is better) — its signature advantage over acid zinc (3:1–5:1).

Titrate / Titration: A chemical test that measures the concentration of a solution so you know whether to add more.

Trivalent: See Cr(III).

Turnover (Tank Turnover): One complete pass of the entire tank volume through the filter (or one full replacement of rinse volume).

Water Break Test: The free, reliable test for clean steel: dip and pull the part; water that **sheets** off = clean, water that **beads** = oil remains, reclean.

White Rust: The white, powdery corrosion of bare zinc (zinc oxide/hydroxide). The whole point of passivation is to delay it.

Zinc (as Zincate, Zn(OH)₄²⁻): The metal you're depositing. In alkaline zinc it's dissolved as the negatively-charged zincate complex at **6–15 g/L** — lower than acid zinc — and is replenished separately (steel anodes don't feed it).

Zincate: The dissolved form of zinc in a caustic bath, Zn(OH)₄²⁻. The complexed ion responsible for alkaline zinc's even, well-distributed deposit.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

★ REMEMBER

Most alkaline zinc defects trace back to a **few root causes**: a drifting NaOH:Zn ratio, carbonate buildup, an additive (purifier/carrier/brightener) out of balance, tramp-metal contamination, or a rinse/bright-dip that let drag-in through. Diagnose with a **Hull Cell** before you start dumping chemistry.

⚠ HEADS-UP

Before changing acid strength, pickle time, or anything that increases metal attack, check whether the part is **>40 HRC** — those changes raise hydrogen-embrittlement risk. When in doubt, ask your supervisor and lean on the relief bake.

STATION 01 — SOAK CLEAN

SYMPTOM	LIKELY CAUSE	FIX
Water break fails (water beads)	Low cleaner concentration or temp	Titrate cleaner; increase temperature
Oily film remains	Floating oil redepositing	Run/install skimmer; dump if heavily contaminated
White residue on parts	Hard-water salts or cleaner residue	Check rinse-water quality (use DI)
Etching / discoloration	Cleaner too aggressive or too hot	Reduce temp; milder formulation
Spotty adhesion later	Uneven cleaning; air pockets	Reposition on rack; add agitation

STATION 02 — RINSE (PRE-ACTIVATION)

SYMPTOM	LIKELY CAUSE	FIX
Conductivity high (>500 µS/cm)	Too much drag-out; low flow	More drain time off cleaner; more flow
Soapy residue	Inadequate rinse time	Extend dwell; add agitation
Acid tank going alkaline	Cleaner carry-over	Improve rinse; add a second stage

STATION 03 — ACID ACTIVATION (PICKLE)

SYMPTOM	LIKELY CAUSE	FIX
Dark oxide remains	Acid too dilute; dip too short	Increase concentration; extend time (check HE risk first)
Pitting / etching	Acid too strong/long/warm	Reduce concentration; shorten dip; add inhibitor
Adhesion fails downstream	Incomplete activation / leftover oxide	Re-check acid strength/time; check alkaline drag-in
Excessive HCl fuming	High conc./temp; poor ventilation	Reduce to minimum effective; check exhaust
Dark smut	High dissolved iron; copper	Dump/recharge acid; find copper source
HE cracking/failures	Over-long pickle; no inhibitor; >40 HRC	Shorten pickle; add inhibitor; relief bake

STATION 04 — RINSE (PRE-PLATE / GUARD)

SYMPTOM	LIKELY CAUSE	FIX
Conductivity high (>200 µS/cm)	Heavy acid drag-out; low flow	More drain off pickle; more flow
Plating bath ratio drifting	Acid drag-in neutralizing caustic	Tighten rinse; add double rinse
Filter loading with sludge fast	Iron dragging in from spent pickle	Dump/refresh pickle; improve rinse
Parts oxidizing before plating	Transfer too slow	Into the plating bath within 30 sec

STATION 05 — ALKALINE ZINC PLATING

SYMPTOM	LIKELY CAUSE	FIX
Dull/dark in low-current areas	Low purifier/carrier; tramp metal (Pb/Cu/Cd)	Add purifier per TDS; analyze metals; dummy; Hull Cell
Dull/matte overall	Low brightener; temp too low	Add brightener per amp-hour; check temp
Burnt/dark at high-current edges	Ratio too low (zinc too high); excess brightener	Adjust ratio; reduce brightener; Hull Cell
Thin / slow plating	Low zinc; ratio too high; low temp	Check zinc/ratio/temp; extend time (low efficiency is normal)
Hazy/cloudy deposit	Carbonate high; organics; excess caustic	Analyze carbonate (chill-treat); carbon treat
Blistering / poor adhesion	Tramp metal; organic breakdown; poor prep	Analyze metals; check prep; reduce brightener
Roughness	Particulate; iron sludge; poor filtration	Check/replace filter; improve filtration
Brittle/cracking deposit	Excess brightener; high stress	Reduce brightener; rebalance per Hull Cell

STATION 06 — RINSE (POST-PLATE / RECOVERY)

SYMPTOM	LIKELY CAUSE	FIX
Bright dip neutralizing fast	Caustic drag-in	Tighten rinse; add/strengthen stage
Chromate thin/blotchy	Residual caustic raising chromate pH	Improve rinse agitation/time
White haze on zinc	Zinc oxidizing; hard water	Use DI; reduce transfer time
Plating bath running lean	Drag-out not recovered	Pump recovery rinse back per schedule

STATION 07 — ACID BRIGHT DIP

SYMPTOM	LIKELY CAUSE	FIX
Zinc still dull after dip	Dip weak/spent or too short	Refresh/strengthen; extend slightly (seconds)
Zinc etched/thin	Dip too strong/long	Dilute; shorten time
Patchy chromate downstream	Uneven bright dip; caustic carry-over	Improve post-plate rinse; refresh bright dip
Brown fumes over tank	Nitric high; poor exhaust	Check ventilation; verify dip strength

STATION 08 — PASSIVATE (TRIVALENT CHROMATE)

SYMPTOM	LIKELY CAUSE	FIX
Rainbow when should be clear	pH too low; time too long	Check pH; reduce time
Pale / thin film	pH too high; bath depleted	Lower pH; replenish per TDS
Blotchy / uneven color	Caustic carry-over; weak bright dip; uneven zinc	Improve rinse & bright dip; check plating
Film rubs off	pH out of range; not enough cure	Check pH; allow 24-hr cure
Bath life short	Caustic/zinc/chloride drag-in	Improve post-plate rinse

HULL CELL QUICK-READ (267 ML • 1–2 A • ~5–10 MIN)

PANEL PATTERN	LIKELY CAUSE
Uniform/bright across panel	Bath in balance
Dull/dark at low-current (thin) end	Low purifier/carrier; tramp metal
Dull/matte overall	Low brightener
Burnt/dark at high-current end	Ratio too low; excess brightener
Skip-plating in low-current area	Strong metallic contamination (Pb/Cu)
Blistering	Organic breakdown or metallic contamination



TIP

Most downstream defects are born upstream. A blotchy chromate is usually a post-plate-rinse or bright-dip problem; a dark recess is usually a purifier problem. Fix the root, not the symptom.

— TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO



REMEMBER

An operator is “line-qualified” only when every station they run is signed. Nobody touches the **rectifier**, the **acid pickle**, the **caustic plating bath**, or the **passivate** solo until the **Safety / PPE & SDS** column is signed. Safety qualification comes first, always.

Sign-off levels: **O** = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Safety / PPE & SDS (all stations)	_____	_____	_____	_____
2	Station 01 — Soak Clean + water break test	_____	_____	_____	_____
3	Station 02 — Rinse (Pre-Act) + conductivity	_____	_____	_____	_____
4	Station 03 — Acid Activate (incl. HE check)	_____	_____	_____	_____
5	Station 04 — Rinse (Pre-Plate / Guard)	_____	_____	_____	_____
6	Station 05 — Alkaline Zinc Plating operation	_____	_____	_____	_____
7	NaOH:Zn ratio control & carbonate awareness	_____	_____	_____	_____
8	Hull Cell / bath test — run & interpret	_____	_____	_____	_____
9	Station 06 — Post-Plate / Recovery Rinse	_____	_____	_____	_____
10	HE Bake (ASTM B850) decision & logging	_____	_____	_____	_____
11	Station 07 — Acid Bright Dip / Activation	_____	_____	_____	_____
12	Station 08 — Passivate (trivalent chromate)	_____	_____	_____	_____
13	Stage 09 — Final Rinse / Dry / Seal & cure	_____	_____	_____	_____
14	LOTO on rectifier	_____	_____	_____	_____
15	Emergency response — eyewash, shower, spill	_____	_____	_____	_____
16	Waste / drag-out / recovery handling	_____	_____	_____	_____
17	Troubleshooting — symptom recognition	_____	_____	_____	_____

Operator name: _____ Hire/start date: _____

Supervisor name: _____ Review cycle: ☐ Annual ☐ Other: _____



HEADS-UP

Safety rows (1, 14, 15) are not optional and not “fill in later.” An operator may not work *any* station until PPE/SDS, LOTO, and emergency response are signed off. Safety competency comes before production competency, always.



TIP

Re-verify each operator annually, and **immediately** after any process or chemistry change. Note the recertification date in the cell (e.g., “Q 6/26 · recert 6/27”). People drift from procedure over time, and a yearly refresher catches bad habits before they become defects — or injuries.

20 QUESTIONS • PASSING SCORE 80% (16/20)

ALKALINE ZINC COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED



REMEMBER

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q5, Q6, Q14, Q19) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: _____ Date: _____ Score: _____ / 20 **Safety-gate Qs (all must be correct): 5, 6, 14, 19**

Q1. Modern cyanide-free alkaline zinc dissolves its zinc in which chemical, which also carries the current?

A. Hydrochloric acid B. Boric acid C. Sodium hydroxide (caustic soda) D. Sodium cyanide

Q2. (Short answer) What is the **water break test**, and what result tells you a part is clean?

Q3. The soak-clean tank typically runs at:

A. 130–160°F B. 70–85°F C. 200–250°F D. Below freezing

Q4. (Short answer) Name the main job of the **acid activation (pickle)** stage. In one sentence, why does the manual say “no activation, no adhesion”?

Q5. [SAFETY GATE] Which station is described as the **single greatest source of hydrogen embrittlement** on the line?

A. Soak clean B. Acid activation (pickle) C. Post-plate rinse D. Passivate

Q6. [SAFETY GATE] A part is **>40 HRC** (high-strength steel). Per ASTM B850, the hydrogen-embrittlement relief bake is:

A. 200°F for 1 hour, any time within a week B. 375 ±25°F for at least 8 hours, started within 4 hours of plating C. 500°F for 30 minutes D. No bake is ever needed for alkaline zinc

Q7. (Short answer) The single most important control number in the alkaline zinc plating bath is the **NaOH:Zn ratio (~10:1)**. In one sentence, what happens to throwing power and plating speed as that ratio goes *up* (more caustic relative to zinc)?

Q8. Compared with acid zinc, alkaline zinc's cathode efficiency is:

A. Higher (95–98%) B. Lower (about 50–80%) C. Exactly 100% D. The same

Q9. (Short answer) Name alkaline zinc's biggest *advantage* over acid zinc, and its biggest *tradeoff*.

Q10. In modern alkaline zinc, the anodes are usually made of:

A. Pure zinc that dissolves to feed the bath B. Copper C. Bare steel (inert); zinc is fed separately D. Graphite

Q11. (Short answer) Name the **three additive types** in a cyanide-free alkaline zinc bath and, in a few words, what each does.

Q12. A Hull Cell panel is **dull/dark in the low-current (thin) area**. The most likely cause is:

A. Low purifier/carrier or tramp-metal contamination B. Too much brightener C. pH too low D. Bath too hot

Q13. Which contaminant does alkaline zinc handle **much better** than acid zinc, because it precipitates out and is filtered away?

A. Lead B. Copper C. Iron D. Cadmium

Q14. (Short answer) [SAFETY GATE] The alkaline zinc *plating bath itself* is one of the most dangerous tanks on the line. What is the chemical hazard it poses (beyond the rectifier), and what is the immediate first aid for skin or eye contact?

Q15. A byproduct that slowly builds up in any caustic bath (from absorbing CO₂ from the air) and must eventually be removed is:

A. Boric acid B. Nitric acid C. Chloride D. Sodium carbonate

Q16. The final passivation station uses which type of chromium in modern RoHS-compliant lines?

A. Hexavalent — Cr(VI) B. No chromium at all C. Cyanide D. Trivalent — Cr(III)

Q17. (Short answer) What is the job of the **acid bright dip (Station 07)** between plating and chromate on an alkaline zinc line?

Q18. A trivalent clear/blue passivate's critical pH window is roughly:

A. 1.5–2.2 B. 4.8–5.8 C. 7.0 D. 11–13



SAFETY SECTION — MUST BE CORRECT TO CERTIFY

Questions 5, 6, 14, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q19. (Short answer / Safety) [SAFETY GATE] List **four** pieces of PPE you must wear when working at the **alkaline zinc plating bath**, and name the procedure you must perform before doing any maintenance on the **rectifier**.

Q20. (Short answer) A customer complains their parts come out "thin and bare deep inside a blind hole." Briefly explain why **alkaline zinc** is a good process choice for that problem, and name the one tradeoff they should expect.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

**HEADS-UP**

Keep this page out of the trainee's copy. Questions **5, 6, 14, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

- Q1. C** — Sodium hydroxide (caustic soda). It dissolves the zinc as zincate *and* carries the current; the modern bath has no cyanide.
- Q2.** Dip and pull the part and watch the water. If it **sheets off in an unbroken, continuous film**, the part is clean. If it **beads up**, oil remains — reclean. (It's free; use it.)
- Q3. A** — 130–160°F (54–71°C); do not exceed 180°F.
- Q4.** It **dissolves the surface oxide/rust** and produces a chemically active steel surface so zinc can bond. Without it, oxide blocks the bond and the plating won't stick (no adhesion).
- Q5. B** — Acid activation (pickle).
- Q6. B** — 375 ±25°F for at least 8 hours, started within 4 hours of plating (ASTM B850).
- Q7.** As the ratio goes *up* (more caustic, leaner zinc), **throwing power improves (more even coverage) but plating gets slower** (lower efficiency). Lower ratio = faster but worse distribution. (Full credit for naming the throwing-power-up / speed-down tradeoff.)
- Q8. B** — Lower, about 50–80% (vs. acid zinc's 95–98%), which is why alkaline zinc plates 2–3× slower.
- Q9. Advantage:** excellent **throwing power / even distribution** (also accept ductile deposits, no cyanide, iron-forgiving). **Tradeoff:** **slower plating / lower efficiency** (also accept hotter/stronger caustic hazard, carbonate buildup, anode-feed management).
- Q10. C** — Bare steel (inert) anodes; zinc is replenished separately (dissolver tank or chemical additions). Some shops still run zinc anodes — accept a note of that.
- Q11. Purifier** — sequesters tramp metals (Pb/Cu/Cd/Fe), keeps low-current areas clean; **Carrier (grain refiner)** — refines grain, supports coverage/throwing power; **Brightener** — adds luster and leveling.
- Q12. A** — Low purifier/carrier or tramp-metal contamination (low-current areas are alkaline zinc's "tell-tale" zone).
- Q13. C** — Iron. It precipitates as iron hydroxide and the filter removes it (acid zinc, by contrast, accumulates iron permanently).
- Q14.** The bath is a **strong (hot) caustic / sodium hydroxide — a severe burn and eye hazard**. First aid: **flush the skin or eyes with water for at least 15 minutes** (longer for caustic, which keeps burning even when it doesn't sting at first) and get medical attention. Accept "caustic/alkaline burn" + "15-minute flush."
- Q15. D** — Sodium carbonate (Na₂CO₃). Removed by chill-crystallization/filtration or partial replacement.
- Q16. D** — Trivalent, Cr(III). (Hexavalent, Cr(VI), is a known carcinogen and not RoHS-compliant.)
- Q17.** It **neutralizes the leftover caustic film and brightens/activates the zinc** so the chromate forms an even, well-colored, well-adhered coating. "Activation, round two — activating the zinc for the chromate." (Accept any of: neutralize caustic, brighten zinc, prep for even chromate.)
- Q18. A** — 1.5–2.2.
- Q19. PPE** — the four **MUST include both sealed chemical splash goggles AND a face shield** (both are mandatory at this hot-caustic tank), plus any two of: chemical-resistant gloves (rated for caustic), chemical apron, chemical-resistant boots, respirator if ventilation is inadequate. Do *not* accept "safety glasses" in place of sealed goggles. **Rectifier procedure: LOTO (Lockout/Tagout)** — de-energize and lock out the power before any rectifier maintenance.
- Q20.** Alkaline zinc has **excellent throwing power** — it plates evenly into deep recesses and blind holes where acid zinc would go thin or bare (surface-to-recess ~1.5:1–2:1 vs acid's 3:1–5:1). **Tradeoff:** it plates **slower (lower efficiency)**, so hitting thickness takes more time. (Accept also: stronger-caustic hazard.)

Answer distribution (MC): A=3 · B=3 · C=3 · D=2

**REMEMBER**

A score of 16/20 passes — but every safety question (5, 6, 14, 19) must be correct to certify. If a trainee misses a safety item, re-teach it and re-test before sign-off. We don't gamble with people.

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CERTIFICATE OF
COMPLETION

ALKALINE ZINC PLATING TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Alkaline Zinc Plating** training course — covering all nine process stations (Clean, Rinse, Activate, Rinse, Plate, Rinse, Bright-Dip/Activate, Passivate, and Final Rinse/Dry), cyanide-free zincate bath chemistry and the NaOH:Zn ratio, additive and carbonate control, Hull Cell diagnostics, hydrogen-embrittlement control and baking, troubleshooting, and the safety practices for alkaline cleaners, the caustic plating bath, mineral acids, the bright dip, and chromate passivation — and has passed the Completion Test with a score of _____ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: _____ Re-certification due: _____

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer specifications, and applicable regulatory requirements before production use. Hexavalent chromium, where used, is a confirmed human carcinogen — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.

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